

BACHELOR IN COMPUTER SCIENCE

DATA AND INFORMATION MANAGEMENT (TEB 1103)

SEMESTER MAY 2026

GROUP 10

PROJECT DATABASE SYSTEM

LECTURER'S NAME :

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1 Company Background

Lumut Naval Shipyard Sdn Bhd (LUNAS) is a Malaysian based defense and maritime enterprise dedicated to providing the high-standard shipbuilding, heavy engineering and maintenance, repair and overhaul (MRO) solutions. Besides that, the company was founded with a commitment to national security and excellence in operation. Based in the strategic Naval Hub of Lumut, Perak, LUNAS has established itself as a stable foundation for both the public and defense sectors since its initial inception in 1984 as the Royal Malaysian Navy Dockyard, eventually transitioning to its fully modernized corporate structure under the ownership of Ocean Sunshine Berhad.

The company is distinguished by a team of highly skilled engineers and specialized technical professionals who can commit in maintaining the highest standards of maritime technology across diverse defense platforms, especially in supporting the Royal Malaysian Navy. Besides traditional ship construction, LUNAS has strategically expended their expertise into specialized engineering divisions, such as the Combat System Centric (CSC) department and the country's exclusive Shore Integration Facility (SIF). Without hesitation, this will be ensured that the cutting-edge of the naval vessels like Littoral Combat Ships (LCS) and Littoral Mission Ships (LMS) are systematically integrated, tested and maintained at the highest performance to provide the smooth defense and operational preparedness.

The sustainability and localized corporate responsibility are the essential components of the company's operational identity. Other than their strong Vendor Development Programme (VDP), LUNAS also actively promotes environmentally responsible practices which are aligned with LUNAS inherent sense of responsibility especially in developing local industrial capabilities, upskilling the Malaysian workforce and driving sustainable socioeconomic development within the local maritime ecosystem. Moreover, the company's versatility is further improved through their specialized subsidiaries such as LUNAS Shiprepair, LUNAS Langkawi and LUNAS Yachts which can help to extend the company's technical capabilities into commercial vessel maintenance, fabrication of heavy steel structures and bespoke maritime engineering tasks.

LUNAS is led by a dynamic management team which is named by Ir. Azhar Jumaat (CEO) and Ir. Azman Shaari (COO) who have the strong academic backgrounds especially in engineering and maritime operations and is supported by extensive hands-on experience in complex naval project coordination. Furthermore, the company is also supported by an experienced board led by Chairman Datuk Dr. Shahrazat Bin Haji Ahmad, which could make LUNAS able to keep continues to build international and domestic trust through customized engineering solutions, uncompromising reliability, and rigid quality control. As the company grows, LUNAS remains focused on scaling its localized defense capabilities while fostering long-term strategic relationships through resilient, which is client-centric maritime delivery.

2 Proof of SSM

SURUHANJAYA SYARIKAT MALAYSIA
COMPANIES COMMISSION OF MALAYSIA

Although all efforts has been carried out to ensure that the information provided is accurate and up to date, the Registrar of Companies will not be liable for any losses arising from any inaccurate or omitted information

CORPORATE INFORMATION

Name	: LUMUT NAVAL SHIPYARD SDN. BHD.
Last Old Name	: BOUSTEAD NAVAL SHIPYARD SDN. BHD.
Date of Change	: 17-01-2024
Company Number	: **199001007732 (199302-K)**
Incorporation Date	: 13-06-1990
Registration Date	: Nil
Type	: LIMITED BY SHARES : PRIVATE LIMITED
Status	: EXISTING
Registered Address	: LEVEL 33, MOF INC. TOWER, 9, PERSIARAN KLCC, KUALA LUMPUR WILAYAH PERSEKUTUAN
Postcode	: 50088
Origin	: MALAYSIA
Business Address	: PENGKALAN TLDM LUMUT PERAK
Postcode	: 32100
Nature of Business	: CONSTRUCTION, REPAIR AND MAINTENANCE OF NAVAL SHIP,WEAPONS AND ELECTRONICS,DESIGN AND ENGINEERING ,FABRICATION OF STEEL STRUCTURES AND COMMERCIAL SHIPBUILDING

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SURUHANJAYA SYARIKAT MALAYSIA
COMPANIES COMMISSION OF MALAYSIA

AKTA SYARIKAT 2016 [Seksyen 28]

No. Pendaftaran : 199001007732 (199302-K)

NOTIS PERAKUAN PEMERBADANAN ATAS PERTUKARAN NAMA SYARIKAT

Dengan ini diperakui bahawa

BOUSTEAD NAVAL SHIPYARD SDN. BHD.

yang telah diperbadankan di bawah Akta Syarikat 2016, pada 13 haribulan Jun 1990, sebagai sebuah syarikat persendirian, pada 17 haribulan Januari 2024 telah menukar namanya kepada

LUMUT NAVAL SHIPYARD SDN. BHD.

dan bahawa syarikat ini adalah sebuah syarikat persendirian dan adalah sebuah syarikat berhad menurut syer.

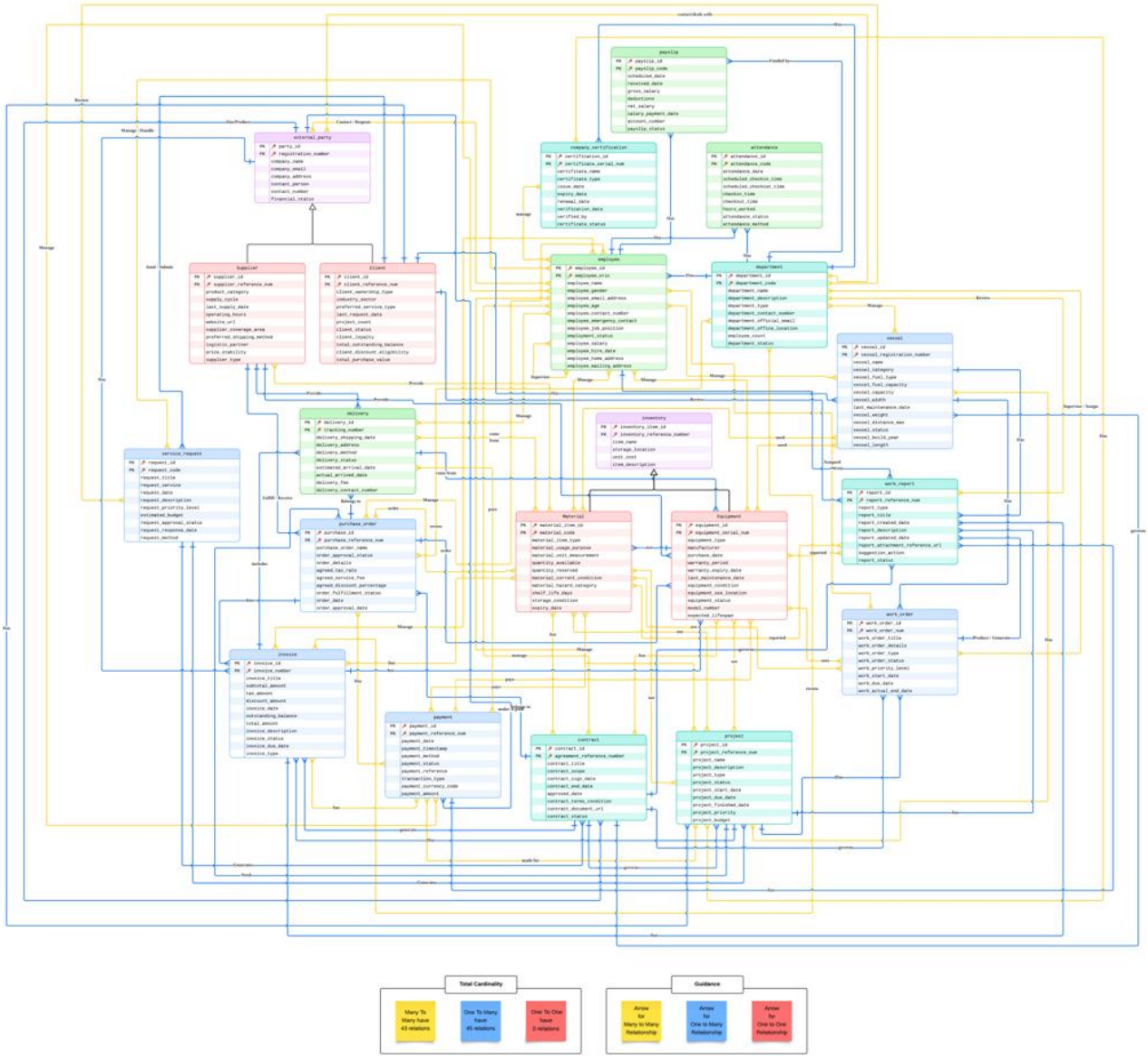
Pendaftar Syarikat Malaysia

CERTIFIED TRUE COPY
NOR FAZIANA BINTI ZAKARIA
Company Secretary
LS 0010646

Notis ini dicetak oleh komputer dan tidak memerlukan tandatangan.
User ID : syati Date : 17/01/2024 13:50:17

MENARA SSM@SENTRAL
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3 Enhanced Entity Relational Diagram (EERD)



4 Part 1

4.1 Relational Schema

1. **Employee**

   { employee_id (PK), employee_nric (PK), employee_name, employee_gender, employee_email_address, employee_home_address, employee_age, employee_contact_number, employee_emergency_contact, employee_job_position, employment_status, employee_salary, employee_hire_date, employee_mailing_address, department_id (FK), department_code (FK) }

2. **Client (Subclass of external_party)**

   { client_id (PK), client_reference_num (PK), client_ownership_type, industry_sector, preferred_service_type, last_request_date, project_count, client_status, client_loyalty, total_outstanding_balance, client_discount_eligibility, total_purchase_value, party_id (FK) , registration_number (FK) }

3. **Supplier (Subclass of external_party)**

   { supplier_id (PK), supplier_reference_num (PK), product_category, supply_cycle, last_supply_date, operating_hours, website_url, supplier_coverage_area, preferred_shipping_method, logistic_partner, price_stability, supplier_type, party_id (FK) , registration_number (FK) }

4. **external_party (Superclass)**

   { party_id (PK), registration_number (PK), company_name, company_email, company_address, contact_person, contact_number, financial_status }

5. **department**

   { department_id (PK), department_code (PK), department_name, department_description, department_type, department_contact_number, department_official_email, department_office_location, employee_count, department_status }

6. **payment**

   { payment_id (PK), payment_reference_num (PK), payment_date, payment_timestamp, payment_method, payment_status, payment_reference, transaction_type, payment_currency_code, payment_amount, party_id (FK), registration_number (FK) }

7. **service_request**

   { request_id (PK), request_code (PK), request_title, request_service, request_date, request_description, request_priority_level, estimated_budget, request_approval_status, request_response_date, request_method, client_id (FK), client_reference_num (FK) }

8. **attendance**

   { attendance_id (PK), attendance_code (PK), attendance_date, scheduled_checkin_time, scheduled_checkout_time, checkin_time, checkout_time, hours_worked, attendance_status, attendance_method, employee_id (FK), employee_nric (FK), department_id (FK), department_code (FK)}

9. **payslip**

   { payslip_id (PK), payslip_code (PK), scheduled_date, received_date, gross_salary, deductions, net_salary, salary_payment_date, account_number, payslip_status, employee_id (FK), employee_nric (FK), department_id (FK), department_code(FK) }

10. **company_certification**

    { certification_id (PK), certificate_serial_num (PK), certificate_name, certificate_type, issue_date, expiry_date, renewal_date, verification_date, verified_by, certificate_status, department_id (FK), department_code(FK) }

11. **work_order (confidential which the client can't access it)**

    { work_order_id (PK), work_order_num (PK), work_order_title, work_order_details, work_order_type, work_order_status, work_priority_level, work_start_date, work_due_date, work_actual_end_date, vessel_id (FK), vessel_registration_number (FK), project_id (FK), project_reference_num (FK), contract_id (FK), agreement_reference_number (FK) }

12. work_report

{ report_id (PK), report_reference_num (PK), report_type, report_title, report_created_date, report_description, report_updated_date, report_attachment_reference_url, suggestion_action, report_status, vessel_id (FK), vessel_registration_number (FK), work_order_id (FK), work_order_num (FK), employee_id (FK), employee_nric (FK), client_id (FK), client_reference_num (FK), contract_id (FK), agreement_reference_number (FK), project_id (FK), project_reference_num (FK), invoice_id (FK), invoice_number (FK), payment_id (FK), payment_reference_num (FK)}

13. contract

{ contract_id (PK), agreement_reference_number (PK), contract_title, contract_scope, contract_sign_date, contract_end_date, approved_date, contract_terms_condition, contract_document_url, contract_status, request_id (FK), request_code (FK), party_id (FK), registration_number (FK)}

14. invoice

{ invoice_id (PK), invoice_number (PK), invoice_title, subtotal_amount, tax_amount, discount_amount, invoice_date, outstanding_balance, total_amount, invoice_description, invoice_status, invoice_due_date, invoice_type, project_id (FK), project_reference_num (FK), contract_id (FK), agreement_reference_number (FK), party_id (FK), registration_number (FK), purchase_id (FK), purchase_reference_num (FK)}

15. project

{ project_id (PK), project_reference_num (PK), project_name, project_description, project_type, project_status, project_start_date, project_due_date, project_finished_date, project_priority, project_budget, contract_id (FK), agreement_reference_number (FK), request_id (FK), request_code (FK), client_id (FK), client_reference_num (FK)}

16. purchase_order

{ purchase_id (PK), purchase_reference_num (PK), purchase_order_name, order_approval_status, order_details, agreed_tax_rate, agreed_service_fee, agreed_discount_percentage, order_fulfillment_status, order_date, order_approval_date, supplier_id (FK), supplier_reference_num (FK), project_id (FK), project_reference_num (FK), contract_id (FK), agreement_reference_number (FK)}

17. vessel

{ vessel_id (PK), vessel_registration_number (PK), vessel_name, vessel_category, vessel_fuel_type, vessel_fuel_capacity, vessel_capacity, vessel_width, last_maintenance_date, vessel_weight, vessel_distance_max, vessel_status, vessel_build_year, vessel_length, client_id (FK), client_reference_num (FK), contract_id (FK), agreement_reference_number (FK)}

18. inventory

{ inventory_item_id (PK), inventory_reference_number (PK), item_name, storage_location, unit_cost, item_description }

19. Material

{ material_item_id (PK), material_code (PK), material_item_type, material_usage_purpose, material_unit_measurement, quantity_available, quantity_reserved, material_current_condition, material_hazard_category, shelf_life_days, storage_condition, expiry_date, inventory_item_id (FK), inventory_reference_number (FK), equipment_id (FK), equipment_serial_num (FK)}

20. Equipment

{ equipment_id (PK), equipment_serial_num (PK), equipment_type, manufacturer, purchase_date, warranty_period, warranty_expiry_date, last_maintenance_date, equipment_condition, equipment_used_location, equipment_status, model_number, expected_lifespan, inventory_item_id (FK), inventory_reference_number (FK), supplier_id (FK), supplier_reference_num (FK), purchase_id (FK), purchase_reference_num (FK), delivery_id (FK), tracking_number (FK), invoice_id (FK), invoice_number (FK)}

21. delivery

{ delivery_id (PK), tracking_number (PK), delivery_shipping_date, delivery_address, delivery_method, delivery_status, estimated_arrival_date, actual_arrived_date, delivery_fee, delivery_contact_number, purchase_id (FK), purchase_reference_num (FK), invoice_id (FK), invoice_number (FK), supplier_id (FK), supplier_reference_num (FK) }

4.2 Relational Schema – Explosion Table

1. employee_supervisor { supervisor employee id , supervisor employee nric, supervisee employee id , supervisee employee nric }
2. invoice_employee { employee id , employee nric , invoice id , invoice number }
3. employee_service_request { employee id , employee nric , request id , request code }
4. employee_delivery { employee id , employee nric , delivery id , tracking number }
5. employee_project { employee id , employee nric , project id , project reference num }
6. contract_employee { employee id , employee nric , contract id , agreement reference number }
7. employee_purchase_order { employee id , employee nric , purchase id , purchase reference num }
8. external_party_employee { employee id, employee nric, party id, registration number }
9. employee_payment { employee id , employee nric , payment id , payment reference num }
10. employee_work_order { employee id , employee nric , work order id , work order num }
11. employee_vessel { employee id , employee nric , vessel id , vessel registration number }
12. employee_company_certification { employee id , employee nric, certification id , certificate serial num }
13. employee_equipment { employee id, employee nric, equipment id, equipment serial num }
14. employee_material { employee id, employee nric, material item id, material code }
15. equipment_project { equipment id, equipment serial num, project id, project reference num}
16. equipment_vessel {equipment id, equipment serial num, vessel id, vessel registration number}
17. equipment_work_order {equipment id, equipment serial num, work order id, work order num}
18. equipment_work_report { equipment id, equipment serial num, report id, report reference num }
19. material_project { material item id, material code, project id, project reference num }
20. material_vessel {material item id, material code, vessel id, vessel registration number}

21. material_work_order {material item id, material code, work order id, work order num}
22. material_work_report {material item id, material code, report id, report reference num}
23. department_service_request { department id , department code , request id , request code }
24. department_work_order { department id , department code , work order id , work order num }
25. department_work_report { department id , department code , report id , report reference num }
26. department_vessel { department id , department code , vessel id , vessel registration number }
27. department_external_party { department id , department code , party id , registration number }
28. department_purchase_order { department id , department code , purchase id , purchase reference num }
29. department_invoice { department id , department code , invoice id , invoice number }
30. material_delivery { material item id, material code , delivery id , tracking number }
31. purchase_order_material { purchase id, purchase reference num, material item id , material code }
32. project_vessel { project id, project reference num, vessel id, vessel registration number }
33. project_company_certification { project id, project reference num, certification id , certificate serial num }
34. invoice_payment { invoice id , invoice number , payment id , payment reference num }
35. project_payment { project id , project reference num , payment id , payment reference num }
36. purchase_order_payment { purchase id , purchase reference num , payment id , payment reference num }
37. supplier_material { supplier id , supplier reference num , material item id , material code }
38. material_contract {material item id, material code, contract id, agreement reference number}
39. payment_material {payment id, payment reference num, material item id, material code}
40. equipment_contract {equipment id, equipment serial num, contract id, agreement reference number}
41. invoice_material {invoice id, invoice number, material item id, material code}

42. payment_equipment {payment_id, payment_reference_num, equipment_id, equipment_serial_num}
43. delivery_payment {delivery_id, tracking_number, payment_id, payment_reference_num}

4.3 Data Specification

1. employee

Attribute	Data Type	Constraints
employee_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL, UNIQUE
employee_nric	VARCHAR2(20)	PRIMARY KEY, NOT NULL, UNIQUE
employee_name	VARCHAR2(255)	NOT NULL, CHECK (LENGTH(employee_name) > 0)
employee_gender	VARCHAR2(10)	CHECK (gender IN ('M', 'F'))
employee_email_address	VARCHAR2(50)	NOT NULL (LENGTH(employee_email_address) > 0)
employee_age	INTEGER	NOT NULL
employee_contact_number	VARCHAR2(30)	NOT NULL
employee_emergency_contact	VARCHAR2(30)	NOT NULL
employee_job_position	VARCHAR2(50)	NOT NULL
employment_status	VARCHAR2(20)	CHECK (employment_status IN ('Active', 'Resigned', 'Suspended', 'On Leave', 'Retired', 'Contract'))
employee_salary	NUMBER(20, 2)	NOT NULL
employee_hire_date	DATE	NOT NULL
employee_home_address	VARCHAR2(1000)	NOT NULL
employee_mailing_address	VARCHAR2(1000)	NOT NULL
department_id	VARCHAR2(16)	FOREIGN KEY REFERENCES department(department_id), NOT NULL
department_code	VARCHAR2(20)	FOREIGN KEY REFERENCES department(department_code), NOT NULL

2. external_party (Superclass)

Attribute	Data Type	Constraints
party_id	VARCHAR2(16)	PRIMARY KEY NOT NULL UNIQUE
registration_number	VARCHAR2(20)	PRIMARY KEY NOT NULL UNIQUE
company_name	VARCHAR2(255)	NOT NULL (LENGTH(company_name) > 0)
company_email	VARCHAR2(255)	NOT NULL (LENGTH(company_email) > 0)
company_address	VARCHAR2(1000)	NOT NULL (LENGTH(company_address) > 0)
contact_person	VARCHAR2(30)	NOT NULL
contact_number	VARCHAR2(30)	NOT NULL
financial_status	VARCHAR2(16)	NOT NULL

3. Client (Subclass of external_party)

Attribute	Data Type	Constraints
client_id	VARCHAR2(16)	PRIMARY KEY NOT NULL UNIQUE
client_reference_num	VARCHAR2(20)	PRIMARY KEY NOT NULL UNIQUE
client_ownership_type	VARCHAR2(25)	NOT NULL
industry_sector	VARCHAR2(30)	NOT NULL
preferred_service_type	VARCHAR2(50)	NOT NULL
last_request_date	DATE	NOT NULL
project_count	INT	NOT NULL
client_status	VARCHAR2(20)	NOT NULL
client_loyalty	VARCHAR2(50)	NOT NULL
total_outstanding_balance	NUMBER(20,2)	NOT NULL
client_discount_eligibility	VARCHAR2(50)	NOT NULL
total_purchase_value	NUMBER(20,2)	NOT NULL
party_id	VARCHAR2(16)	FOREIGN KEY REFERENCES external_party(party_id), NOT NULL
registration_number	VARCHAR2(20)	FOREIGN KEY REFERENCES external_party(registration_number), NOT NULL

4. Supplier (Subclass of external_party)

Attribute	Data Type	Constraints
supplier_id	VARCHAR2(16)	PRIMARY KEY NOT NULL UNIQUE
supplier_reference_num	VARCHAR2(20)	PRIMARY KEY NOT NULL UNIQUE
product_category	VARCHAR2(50)	NOT NULL
supply_cycle	VARCHAR2(50)	NOT NULL
last_supply_date	DATE	NOT NULL
operating_hours	DATE	NOT NULL
website_url	VARCHAR2(255)	NOT NULL
supplier_coverage_area	VARCHAR2(50)	NOT NULL
preferred_shipping_method	VARCHAR2(50)	NOT NULL
logistic_partner	VARCHAR2(50)	NOT NULL
price_stability	VARCHAR2(20)	NOT NULL
supplier_type	VARCHAR2(50)	NOT NULL
party_id	VARCHAR2(16)	FOREIGN KEY REFERENCES external_party(party_id), NOT NULL
registration_number	VARCHAR2(20)	FOREIGN KEY REFERENCES external_party(registration_number), NOT NULL

5. department

Attribute	Data Type	Constraints
department_id	VARCHAR2(16)	PRIMARY KEY UNIQUE NOT NULL
department_code	VARCHAR2(20)	PRIMARY KEY UNIQUE NOT NULL
department_name	VARCHAR2(50)	NOT NULL
department_description	VARCHAR2(255)	NOT NULL
department_type	VARCHAR2(50)	NOT NULL
department_contact_number	VARCHAR2(25)	UNIQUE NOT NULL
department_official_email	VARCHAR2(50)	UNIQUE NOT NULL
department_office_location	VARCHAR2(100)	NOT NULL
employee_count	INTEGER	NOT NULL
department_status	VARCHAR2(20)	NOT NULL

6. payment

Attribute	Data Type	Constraints
payment_id	VARCHAR2(16)	PRIMARY KEY UNIQUE NOT NULL
payment_reference_num	VARCHAR2(20)	PRIMARY KEY UNIQUE NOT NULL
payment_date	DATE	NOT NULL
payment_timestamp	TIME	NOT NULL
payment_method	VARCHAR2(50)	NOT NULL
payment_status	VARCHAR2(50)	NOT NULL
payment_reference	VARCHAR2(255)	NOT NULL
transaction_type	VARCHAR2(100)	NOT NULL
payment_currency_code	VARCHAR2(10)	NOT NULL
payment_amount	NUMBER(20,2)	NOT NULL
party_id	VARCHAR2(16)	FOREIGN KEY REFERENCES external_party(party_id), NOT NULL
registration_number	VARCHAR2(20)	FOREIGN KEY REFERENCES external_party(registration_number), NOT NULL

7. payslip

Attribute	Data Type	Constraints
payslip_id	VARCHAR2(16)	PRIMARY KEY NOT NULL UNIQUE
payslip_code	VARCHAR2(20)	PRIMARY KEY NOT NULL UNIQUE
scheduled_date	DATE	NOT NULL
received_date	DATE	NOT NULL
gross_salary	NUMBER(10,2)	NOT NULL
deductions	NUMBER(10,2)	NOT NULL
net_salary	NUMBER(10,2)	NOT NULL
salary_payment_date	DATE	NOT NULL
account_number	VARCHAR2(50)	NOT NULL
payslip_status	VARCHAR2(30)	NOT NULL
employee_id	VARCHAR2(16)	FOREIGN KEY REFERENCES employee(employee_id)
employee_nric	VARCHAR2(20)	FOREIGN KEY REFERENCES employee(employee_nric)
department_id	VARCHAR2(16)	FOREIGN KEY REFERENCES department(department_id)
department_code	VARCHAR2(20)	FOREIGN KEY REFERENCES department(department_code)

8. service_request

Attribute	Data Type	Constraints
request_id	VARCHAR2(16)	PRIMARY KEY NOT NULL UNIQUE
request_code	VARCHAR2(20)	PRIMARY KEY NOT NULL UNIQUE
request_title	VARCHAR2(255)	NOT NULL
request_service	VARCHAR2(255)	NOT NULL
request_date	DATE	NOT NULL
request_description	VARCHAR2(1000)	NOT NULL
request_priority_level	VARCHAR2(20)	NOT NULL
estimated_budget	NUMBER (20, 2)	NOT NULL
request_approval_status	VARCHAR2(20)	NOT NULL
request_response_date	DATE	NOT NULL
request_method	VARCHAR2(50)	NOT NULL
client_id	VARCHAR2(16)	FOREIGN KEY REFERENCES Client(client_id), NOT NULL
client_reference_num	VARCHAR2(20)	FOREIGN KEY REFERENCES Client(client_reference_num), NOT NULL

9. attendance

Attribute	Data Type	Constraints
attendance_id	VARCHAR2(16)	PRIMARY KEY NOT NULL UNIQUE
attendance_code	VARCHAR2(20)	PRIMARY KEY NOT NULL UNIQUE
attendance_date	DATE	NOT NULL
scheduled_checkin_time	TIME	NOT NULL
scheduled_checkout_time	TIME	NOT NULL
checkin_time	TIME	NOT NULL
checkout_time	TIME	NOT NULL
hours_worked	INTEGER	NOT NULL
attendance_status	VARCHAR2(30)	NOT NULL
attendance_method	VARCHAR2(30)	NOT NULL
employee_id	VARCHAR2(16)	FOREIGN KEY REFERENCES employee(employee_id)
employee_nric	VARCHAR2(20)	FOREIGN KEY REFERENCES employee(employee_nric)
department_id	VARCHAR2(16)	FOREIGN KEY REFERENCES employee(department_id)
department_code	VARCHAR2(20)	FOREIGN KEY REFERENCES employee(department_code)

10. company_certification

Attribute	Data Type	Constraints
certification_id	VARCHAR2(16)	PRIMARY KEY NOT NULL UNIQUE
certificate_serial_num	VARCHAR2(20)	PRIMARY KEY NOT NULL UNIQUE
certificate_name	VARCHAR2(255)	NOT NULL
certificate_type	VARCHAR2(50)	NOT NULL
issue_date	DATE	NOT NULL
expiry_date	DATE	NOT NULL
renewal_date	DATE	NOT NULL
verification_date	DATE	NOT NULL
verified_by	VARCHAR2(50)	NOT NULL
certificate_status	VARCHAR2(30)	NOT NULL
department_id	VARCHAR2(16)	FOREIGN KEY REFERENCES department(department_id)
department_code	VARCHAR2(20)	FOREIGN KEY REFERENCES department(department_code)

11. work_order

Attribute	Data Type	Constraints
work_order_id	VARCHAR2(16)	PRIMARY KEY NOT NULL UNIQUE
work_order_num	VARCHAR2(20)	PRIMARY KEY NOT NULL UNIQUE
work_order_title	VARCHAR2(255)	NOT NULL
work_order_details	VARCHAR2(1000)	NOT NULL
work_order_type	VARCHAR2(255)	NOT NULL
work_order_status	VARCHAR2(20)	NOT NULL
work_priority_level	VARCHAR2(30)	NOT NULL
work_start_date	DATE	NOT NULL
work_due_date	DATE	NOT NULL
work_actual_end_date	DATE	NOT NULL
vessel_id	VARCHAR2(16)	FOREIGN KEY REFERENCES vessel(vessel_id)
vessel_registration_number	VARCHAR2(20)	FOREIGN KEY REFERENCES vessel(vessel_registration_number)
project_id	VARCHAR2(16)	FOREIGN KEY REFERENCES project(project_id)
project_reference_num	VARCHAR2(20)	FOREIGN KEY REFERENCES project(project_reference_num)
contract_id	VARCHAR2(16)	FOREIGN KEY REFERENCES contract(contract_id)
agreement_reference_number	VARCHAR2(20)	FOREIGN KEY REFERENCES contract(agreement_reference_number)

12. work_report

Attribute	Data Type	Constraints
report_id	VARCHAR2(16)	PRIMARY KEY NOT NULL UNIQUE
report_reference_num	VARCHAR2(20)	PRIMARY KEY NOT NULL UNIQUE
report_type	VARCHAR2(255)	NOT NULL
report_title	VARCHAR2(255)	NOT NULL
report_created_date	DATE	NOT NULL
report_description	VARCHAR2(1000)	NOT NULL
report_updated_date	DATE	NOT NULL
report_attachment_reference_url	VARCHAR2(255)	NOT NULL
suggestion_action	VARCHAR2(255)	NOT NULL
report_status	VARCHAR2(30)	NOT NULL
vessel_id	VARCHAR2(16)	FOREIGN KEY REFERENCES vessel(vessel_id)
vessel_registration_number	VARCHAR2(20)	FOREIGN KEY REFERENCES vessel(vessel_registration_number)
work_order_id	VARCHAR2(16)	FOREIGN KEY REFERENCES work_order(work_order_id)
work_order_num	VARCHAR2(20)	FOREIGN KEY REFERENCES work_order(work_order_num)
employee_id	VARCHAR2(16)	FOREIGN KEY REFERENCES employee(employee_id)
employee_nric	VARCHAR2(20)	FOREIGN KEY REFERENCES employee(employee_nric)
client_id	VARCHAR2(16)	FOREIGN KEY REFERENCES Client(client_id)
client_reference_num	VARCHAR2(20)	FOREIGN KEY REFERENCES Client(client_reference_num)
contract_id	VARCHAR2(16)	FOREIGN KEY REFERENCES contract(contract_id)
agreement_reference_number	VARCHAR2(20)	FOREIGN KEY REFERENCES contract(agreement_reference_number)
project_id	VARCHAR2(16)	FOREIGN KEY REFERENCES project(project_id)

project_reference_num	VARCHAR2(20)	FOREIGN KEY REFERENCES project(project_reference_num)
invoice_id	VARCHAR2(16)	FOREIGN KEY REFERENCES invoice(invoice_id)
invoice_number	VARCHAR2(20)	FOREIGN KEY REFERENCES invoice(invoice_number)
payment_id	VARCHAR2(16)	FOREIGN KEY REFERENCES payment(payment_id)
payment_reference_num	VARCHAR2(20)	FOREIGN KEY REFERENCES payment(payment_reference_num)

13. contract

Attribute	Data Type	Constraints
contract_id	VARCHAR2(16)	PRIMARY KEY NOT NULL UNIQUE
agreement_reference_number	VARCHAR2(20)	PRIMARY KEY NOT NULL UNIQUE
contract_title	VARCHAR2(255)	NOT NULL
contract_scope	VARCHAR2(128)	NOT NULL
contract_sign_date	DATE	NOT NULL
contract_end_date	DATE	NOT NULL
approved_date	DATE	NOT NULL
contract_terms_condition	VARCHAR2(1000)	NOT NULL
contract_document_url	VARCHAR2(255)	NOT NULL
contract_status	VARCHAR2(255)	NOT NULL
request_id	VARCHAR2(16)	FOREIGN KEY REFERENCES service_request(request_id)
request_code	VARCHAR2(20)	FOREIGN KEY REFERENCES service_request(request_code)
party_id	VARCHAR2(16)	FOREIGN KEY REFERENCES external_party(party_id)
registration_number	VARCHAR2(20)	FOREIGN KEY REFERENCES external_party(registration_number)

14. invoice

Attribute	Data Type	Constraints
invoice_id	VARCHAR2(16)	PRIMARY KEY NOT NULL UNIQUE
invoice_number	VARCHAR2(20)	PRIMARY KEY NOT NULL UNIQUE
invoice_title	VARCHAR2(255)	NOT NULL
subtotal_amount	NUMBER(20,2)	NOT NULL
tax_amount	NUMBER(20,2)	NOT NULL
discount_amount	NUMBER(20,2)	NOT NULL
invoice_date	DATE	NOT NULL
outstanding_balance	NUMBER(20,2)	NOT NULL
total_amount	NUMBER(20,2)	NOT NULL
invoice_description	VARCHAR2(1000)	NOT NULL
invoice_status	VARCHAR2(30)	NOT NULL
invoice_due_date	DATE	NOT NULL
invoice_type	VARCHAR2(50)	NOT NULL
project_id	VARCHAR2(16)	FOREIGN KEY REFERENCES project(project_id)
project_reference_num	VARCHAR2(20)	FOREIGN KEY REFERENCES project(project_id)
contract_id	VARCHAR2(16)	FOREIGN KEY REFERENCES contract(contract_id)
agreement_reference_number	VARCHAR2(20)	FOREIGN KEY REFERENCES contract(agreement_reference_number)
party_id	VARCHAR2(16)	FOREIGN KEY REFERENCES external_party(party_id)
registration_number	VARCHAR2(20)	FOREIGN KEY REFERENCES external_party(registration_number)
purchase_id	VARCHAR2(16)	FOREIGN KEY REFERENCES purchase_order(purchase_id)
purchase_reference_num	VARCHAR2(20)	FOREIGN KEY REFERENCES purchase_order(purchase_reference_num)

15. project

Attribute	Data Type	Constraints
project_id	VARCHAR2(16)	PRIMARY KEY NOT NULL UNIQUE
project_reference_num	VARCHAR2(20)	PRIMARY KEY NOT NULL UNIQUE
project_name	VARCHAR2(255)	NOT NULL
project_description	VARCHAR2(1000)	NOT NULL
project_type	VARCHAR2(50)	NOT NULL
project_status	VARCHAR2(30)	NOT NULL
project_start_date	DATE	NOT NULL
project_due_date	DATE	NOT NULL
project_finished_date	DATE	
project_priority	VARCHAR2(50)	NOT NULL
project_budget	NUMBER(20,2)	NOT NULL
contract_id	VARCHAR2(16)	FOREIGN KEY REFERENCES contract(contract_id)
agreement_reference_number	VARCHAR2(20)	FOREIGN KEY REFERENCES contract(agreement_reference_number)
request_id	VARCHAR2(16)	FOREIGN KEY REFERENCES service_request(request_id)
request_code	VARCHAR2(20)	FOREIGN KEY REFERENCES service_request(request_code)
client_id	VARCHAR2(16)	FOREIGN KEY REFERENCES Client(client_id)
client_reference_num	VARCHAR2(20)	FOREIGN KEY REFERENCES Client(client_reference_num)

16. purchase_order

Attribute	Data Type	Constraints
purchase_id	VARCHAR2(16)	PRIMARY KEY NOT NULL UNIQUE
purchase_reference_num	VARCHAR2(20)	PRIMARY KEY NOT NULL UNIQUE
purchase_order_name	VARCHAR2(255)	NOT NULL
order_approval_status	VARCHAR2(30)	NOT NULL
order_details	VARCHAR2(1000)	NOT NULL
agreed_tax_rate	NUMBER(20,2)	NOT NULL
agreed_service_fee	NUMBER(20,2)	NOT NULL
agreed_discount_percentage	NUMBER(5,2)	NOT NULL
order_fulfillment_status	VARCHAR2(30)	NOT NULL
order_date	DATE	NOT NULL
order_approval_date	DATE	
supplier_id	VARCHAR2(16)	FOREIGN KEY REFERENCES Supplier(supplier_id)
supplier_reference_num	VARCHAR2(20)	FOREIGN KEY REFERENCES Supplier(supplier_reference_num)
project_id	VARCHAR2(16)	FOREIGN KEY REFERENCES project(project_id)
project_reference_num	VARCHAR2(20)	FOREIGN KEY REFERENCES project(project_reference_num)
contract_id	VARCHAR2(16)	FOREIGN KEY REFERENCES contract(contract_id)
agreement_reference_number	VARCHAR2(20)	FOREIGNKEY REFERENCES contract(agreement_reference_number)

17. vessel

Attribute	Data Type	Constraints
vessel_id	VARCHAR2(16)	PRIMARY KEY NOT NULL UNIQUE
vessel_registration_number	VARCHAR2(20)	PRIMARY KEY NOT NULL UNIQUE
vessel_name	VARCHAR2(255)	
vessel_category	VARCHAR2(50)	NOT NULL
vessel_fuel_type	VARCHAR2(20)	NOT NULL
vessel_fuel_capacity	NUMBER(10,2)	NOT NULL
vessel_capacity	NUMBER	NOT NULL
vessel_width	NUMBER(10,2)	NOT NULL
last_maintenance_date	DATE	NOT NULL
vessel_weight	NUMBER(20,2)	NOT NULL
vessel_distance_max	NUMBER(20,2)	NOT NULL
vessel_status	VARCHAR2(30)	NOT NULL
vessel_build_year	NUMBER	NOT NULL
vessel_length	NUMBER(10,2)	NOT NULL
client_id	VARCHAR2(16)	FOREIGN KEY REFERENCES Client(client_id) NOT NULL
client_reference_num	VARCHAR2(20)	FOREIGN KEY REFERENCES Client(client_reference_num) NOT NULL
contract_id	VARCHAR2(16)	FOREIGN KEY REFERENCES contract(contract_id)
agreement_reference_number	VARCHAR2(20)	FOREIGNKEY REFERENCES contract(agreement_reference_number)

18. inventory (Superclass)

Attribute	Data Type	Constraints
inventory_item_id	VARCHAR2(16)	PRIMARY KEY NOT NULL UNIQUE
inventory_reference_number	VARCHAR2(20)	PRIMARY KEY NOT NULL UNIQUE
item_name	VARCHAR2(255)	NOT NULL
storage_location	VARCHAR2(30)	NOT NULL
unit_cost	NUMBER(20,2)	NOT NULL
item_description	VARCHAR2(1000)	NOT NULL

19. Material (Subclass of inventory)

Attribute	Data Type	Constraints
material_item_id	VARCHAR2(16)	PRIMARY KEY NOT NULL UNIQUE
material_code	VARCHAR2(20)	PRIMARY KEY NOT NULL UNIQUE
material_item_type	VARCHAR2(255)	NOT NULL
material_usage_purpose	VARCHAR2(255)	NOT NULL
material_unit_measurement	VARCHAR2(30)	NOT NULL
quantity_available	INTEGER	NOT NULL
quantity_reserved	INTEGER	NOT NULL
material_current_condition	VARCHAR2(255)	NOT NULL
material_hazard_category	VARCHAR2(255)	NOT NULL
shelf_life_days	INTEGER	NOT NULL
storage_condition	VARCHAR2(50)	NOT NULL
expiry_date	DATE	NOT NULL
equipment_id	VARCHAR2(16)	FOREIGN KEY REFERENCES equipment(equipment_id)
equipment_serial_num	VARCHAR2(20)	FOREIGN KEY REFERENCES equipment(equipment_serial_num)
inventory_item_id	VARCHAR2(16)	FOREIGN KEY REFERENCES inventory(inventory_item_id)
inventory_reference_number	VARCHAR2(20)	FOREIGN KEY REFERENCES inventory(inventory_reference_number)

20. Equipment (Subclass of inventory)

Attribute	Data Type	Constraints
equipment_id	VARCHAR2(16)	PRIMARY KEY NOT NULL UNIQUE
equipment_serial_num	VARCHAR2(20)	PRIMARY KEY NOT NULL UNIQUE
equipment_type	VARCHAR2(50)	NOT NULL
manufacturer	VARCHAR2(100)	NOT NULL
purchase_date	DATE	NOT NULL
warranty_period	VARCHAR2(30)	NOT NULL
warranty_expiry_date	DATE	NOT NULL
last_maintenance_date	DATE	NOT NULL
equipment_condition	VARCHAR2(255)	NOT NULL
equipment_used_location	VARCHAR2(50)	NOT NULL
equipment_status	VARCHAR2(255)	NOT NULL
model_number	VARCHAR2(50)	NOT NULL
expected_lifespan	VARCHAR2(50)	NOT NULL
supplier_id	VARCHAR2(16)	FOREIGN KEY REFERENCES Supplier(supplier_id)
supplier_reference_num	VARCHAR2(20)	FOREIGN KEY REFERENCES Supplier(supplier_reference_num)
purchase_id	VARCHAR2(16)	FOREIGN KEY REFERENCES purchase_order(purchase_id)
purchase_reference_num	VARCHAR2(20)	FOREIGN KEY REFERENCES purchase_order(purchase_reference_num)
delivery_id	VARCHAR2(16)	FOREIGN KEY REFERENCES delivery(delivery_id)
tracking_number	VARCHAR2(20)	FOREIGN KEY REFERENCES delivery(tracking_number)
invoice_id	VARCHAR2(16)	FOREIGN KEY REFERENCES invoice(invoice_id)
invoice_number	VARCHAR2(20)	FOREIGN KEY REFERENCES invoice(invoice_number)
inventory_item_id	VARCHAR2(16)	FOREIGN KEY REFERENCES inventory(inventory_item_id)

inventory_reference_number	VARCHAR2(20)	FOREIGN KEY REFERENCES inventory(inventory_reference_number)

21. delivery

Attribute	Data Type	Constraints
delivery_id	VARCHAR2(16)	PRIMARY KEY NOT NULL UNIQUE
tracking_number	VARCHAR2(20)	PRIMARY KEY NOT NULL UNIQUE
delivery_shipping_date	DATE	NOT NULL
delivery_address	VARCHAR2(1000)	NOT NULL
delivery_method	VARCHAR2(50)	NOT NULL
delivery_status	VARCHAR2(30)	NOT NULL
estimated_arrival_date	DATE	NOT NULL
actual_arrived_date	DATE	
delivery_fee	NUMBER(20,2)	NOT NULL
delivery_contact_number	VARCHAR2(30)	NOT NULL
purchase_id	VARCHAR2(16)	FOREIGN KEY REFERENCES purchase_order(purchase_id)
purchase_reference_num	VARCHAR2(20)	FOREIGN KEY REFERENCES purchase_order(purchase_reference_num)
invoice_id	VARCHAR2(16)	FOREIGN KEY REFERENCES invoice(invoice_id)
invoice_number	VARCHAR2(20)	FOREIGN KEY REFERENCES invoice(invoice_number)
supplier_id	VARCHAR2(16)	FOREIGN KEY REFERENCES Supplier(supplier_id)
supplier_reference_num	VARCHAR2(20)	FOREIGN KEY REFERENCES Supplier(supplier_reference_num)

4.4 Data Specification – Explosion Table

1. employee_supervisor

Attribute	Data Type	Constraints
supervisor_employee_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
supervisor_employee_nric	VARCHAR2(20)	PRIMARY KEY, NOT NULL
supervisee_employee_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
supervisee_employee_nric	VARCHAR2(20)	PRIMARY KEY, NOT NULL

2. **external_party_employee**

Attribute	Data Type	Constraints
employee_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
employee_nric	VARCHAR2(20)	PRIMARY KEY, NOT NULL
party_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
registration_number	VARCHAR2(20)	PRIMARY KEY, NOT NULL

3. invoice_employee

Attribute	Data Type	Constraints
employee_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
employee_nric	VARCHAR2(20)	PRIMARY KEY, NOT NULL
invoice_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
invoice_number	VARCHAR2(20)	PRIMARY KEY, NOT NULL

4. employee_service_request

Attribute	Data Type	Constraints
employee_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
employee_nric	VARCHAR2(20)	PRIMARY KEY, NOT NULL
request_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
request_code	VARCHAR2(20)	PRIMARY KEY, NOT NULL

5. employee_delivery

Attribute	Data Type	Constraints
employee_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
employee_nric	VARCHAR2(20)	PRIMARY KEY, NOT NULL
delivery_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
tracking_number	VARCHAR2(20)	PRIMARY KEY, NOT NULL

6. contract_employee

Attribute	Data Type	Constraints
employee_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
employee_nric	VARCHAR2(20)	PRIMARY KEY, NOT NULL
contract_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
agreement_reference_number	VARCHAR2(20)	PRIMARY KEY, NOT NULL

7. employee_project

Attribute	Data Type	Constraints
employee_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
employee_nric	VARCHAR2(20)	PRIMARY KEY, NOT NULL
project_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
project_reference_num	VARCHAR2(20)	PRIMARY KEY, NOT NULL

8. employee_company_certification

Attribute	Data Type	Constraints
employee_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
employee_nric	VARCHAR2(20)	PRIMARY KEY, NOT NULL
certification_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
certificate_serial_num	VARCHAR2(20)	PRIMARY KEY, NOT NULL

9. work_order_employee

Attributes	Data Type	Constraints
employee_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
employee_nric	VARCHAR2(20)	PRIMARY KEY, NOT NULL
work_order_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
work_order_num	VARCHAR2(20)	PRIMARY KEY, NOT NULL

10. vessel_employee

Attributes	Data Type	Constraints
employee_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
employee_nric	VARCHAR2(20)	PRIMARY KEY, NOT NULL
vessel_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
vessel_registration_number	VARCHAR2(20)	PRIMARY KEY, NOT NULL

11. employee_purchase_order

Attributes	Data Type	Constraints
employee_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
employee_nric	VARCHAR2(20)	PRIMARY KEY, NOT NULL
purchase_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
purchase_reference_num	VARCHAR2(20)	PRIMARY KEY, NOT NULL

12. employee_equipment

Attributes	Data Type	Constraints
employee_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
employee_nric	VARCHAR2(20)	PRIMARY KEY, NOT NULL
equipment_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
equipment_serial_num	VARCHAR2(20)	PRIMARY KEY, NOT NULL

13. employee_payment

Attributes	Data Type	Constraints
employee_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
employee_nric	VARCHAR2(20)	PRIMARY KEY, NOT NULL
payment_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
payment_reference_num	VARCHAR2(20)	PRIMARY KEY, NOT NULL

14. employee_material

Attributes	Data Type	Constraints
employee_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
employee_nric	VARCHAR2(20)	PRIMARY KEY, NOT NULL
material_item_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
material_code	VARCHAR2(20)	PRIMARY KEY, NOT NULL

15. equipment_project

Attributes	Data Type	Constraints
employee_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
employee_nric	VARCHAR2(20)	PRIMARY KEY, NOT NULL
project_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
project_reference_num	VARCHAR2(20)	PRIMARY KEY, NOT NULL

16. material_project

Attributes	Data Type	Constraints
material_item_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
material_code	VARCHAR2(20)	PRIMARY KEY, NOT NULL
project_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
project_reference_num	VARCHAR2(20)	PRIMARY KEY, NOT NULL

17. equipment_vessel

Attributes	Data Type	Constraints
equipment_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
equipment_serial_num	VARCHAR2(20)	PRIMARY KEY, NOT NULL
vessel_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
vessel_registration_number	VARCHAR2(20)	PRIMARY KEY, NOT NULL

18. material_vessel

Attributes	Data Type	Constraints
material_item_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
material_code	VARCHAR2(20)	PRIMARY KEY, NOT NULL
vessel_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
vessel_registration_number	VARCHAR2(20)	PRIMARY KEY, NOT NULL

19. equipment_work_order

Attributes	Data Type	Constraints
equipment_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
equipment_serial_num	VARCHAR2(20)	PRIMARY KEY, NOT NULL
work_order_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
work_order_num	VARCHAR2(20)	PRIMARY KEY, NOT NULL

20. material_work_order

Attributes	Data Type	Constraints
material_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
material_code	VARCHAR2(20)	PRIMARY KEY, NOT NULL
work_order_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
work_order_num	VARCHAR2(20)	PRIMARY KEY, NOT NULL

21. equipment_work_report

Attributes	Data Type	Constraints
equipment_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
equipment_serial_num	VARCHAR2(20)	PRIMARY KEY, NOT NULL
report_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
report_reference_num	VARCHAR2(20)	PRIMARY KEY, NOT NULL

22. material_work_report

Attributes	Data Type	Constraints
material_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
material_code	VARCHAR2(20)	PRIMARY KEY, NOT NULL
report_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
report_reference_num	VARCHAR2(20)	PRIMARY KEY, NOT NULL

23. external_party_department

Attributes	Data Type	Constraints
department_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
department_code	VARCHAR2(20)	PRIMARY KEY, NOT NULL
party_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
registration_number	VARCHAR2(20)	PRIMARY KEY, NOT NULL

24. department_service_request

Attributes	Data Type	Constraints
department_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
department_code	VARCHAR2(20)	PRIMARY KEY, NOT NULL
request_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
request_code	VARCHAR2(20)	PRIMARY KEY, NOT NULL

25. department_work_order

Attributes	Data Type	Constraints
department_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
department_code	VARCHAR2(20)	PRIMARY KEY, NOT NULL
work_order_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
work_order_num	VARCHAR2(20)	PRIMARY KEY, NOT NULL

26. department_work_report

Attributes	Data Type	Constraints
department_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
department_code	VARCHAR2(20)	PRIMARY KEY, NOT NULL
report_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
report_reference_num	VARCHAR2(20)	PRIMARY KEY, NOT NULL

27. department_vessel

Attributes	Data Type	Constraints
department_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
department_code	VARCHAR2(20)	PRIMARY KEY, NOT NULL
vessel_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
vessel_registration_number	VARCHAR2(20)	PRIMARY KEY, NOT NULL

28. purchase_order_department

Attributes	Data Type	Constraints
department_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
department_code	VARCHAR2(20)	PRIMARY KEY, NOT NULL
purchase_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
purchase_reference_num	VARCHAR2(20)	PRIMARY KEY, NOT NULL

29. invoice_department

Attributes	Data Type	Constraints
department_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
department_code	VARCHAR2(20)	PRIMARY KEY, NOT NULL
invoice_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
invoice_number	VARCHAR2(20)	PRIMARY KEY, NOT NULL

30. supplier_material

Attributes	Data Type	Constraints
supplier_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
supplier_reference_num	VARCHAR2(20)	PRIMARY KEY, NOT NULL
material_item_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
material_code	VARCHAR2(20)	PRIMARY KEY, NOT NULL

31. project_vessel

Attributes	Data Type	Constraints
project_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
project_reference_num	VARCHAR2(20)	PRIMARY KEY, NOT NULL
vessel_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
vessel_registration_number	VARCHAR2(20)	PRIMARY KEY, NOT NULL

32. payment_project

Attributes	Data Type	Constraints
project_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
project_reference_num	VARCHAR2(20)	PRIMARY KEY, NOT NULL
payment_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
payment_reference_num	VARCHAR2(20)	PRIMARY KEY, NOT NULL

33. project_company_certification

Attributes	Data Type	Constraints
project_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
project_reference_num	VARCHAR2(20)	PRIMARY KEY, NOT NULL
certification_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
certificate_serial_num	VARCHAR2(20)	PRIMARY KEY, NOT NULL

34. material_delivery

Attributes	Data Type	Constraints
material_item_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
material_code	VARCHAR2(20)	PRIMARY KEY, NOT NULL
delivery_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
tracking_number	VARCHAR2(20)	PRIMARY KEY, NOT NULL

35. purchase_order_material

Attributes	Data Type	Constraints
purchase_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
purchase_reference_num	VARCHAR2(20)	PRIMARY KEY, NOT NULL
material_item_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
material_code	VARCHAR2(20)	PRIMARY KEY, NOT NULL

36. purchase_order_payment

Attributes	Data Type	Constraints
purchase_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
purchase_reference_num	VARCHAR2(20)	PRIMARY KEY, NOT NULL
payment_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
payment_reference_num	VARCHAR2(20)	PRIMARY KEY, NOT NULL

37. invoice_payment

Attributes	Data Type	Constraints
invoice_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
invoice_number	VARCHAR2(20)	PRIMARY KEY, NOT NULL
payment_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
payment_reference_num	VARCHAR2(20)	PRIMARY KEY, NOT NULL

38. material_contract

Attributes	Data Type	Constraints
material_item_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
material_code	VARCHAR2(20)	PRIMARY KEY, NOT NULL
contract_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
agreement_reference_number	VARCHAR2(20)	PRIMARY KEY, NOT NULL

39. payment_material

Attributes	Data Type	Constraints
payment_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
payment_reference_num	VARCHAR2(20)	PRIMARY KEY, NOT NULL
material_item_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
material_code	VARCHAR2(20)	PRIMARY KEY, NOT NULL

40. equipment_contract

Attributes	Data Type	Constraints
equipment_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
equipment_reference_num	VARCHAR2(20)	PRIMARY KEY, NOT NULL
contract_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
agreement_reference_number	VARCHAR2(20)	PRIMARY KEY, NOT NULL

41. invoice_material

Attributes	Data Type	Constraints
invoice_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
invoice_number	VARCHAR2(20)	PRIMARY KEY, NOT NULL
material_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
material_code	VARCHAR2(20)	PRIMARY KEY, NOT NULL

42. payment_equipment

Attributes	Data Type	Constraints
payment_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
payment_reference_num	VARCHAR2(20)	PRIMARY KEY, NOT NULL
equipment_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
equipment_reference_ num	VARCHAR2(20)	PRIMARY KEY, NOT NULL

43. delivery_payment

Attributes	Data Type	Constraints
delivery_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
tracking_number	VARCHAR2(20)	PRIMARY KEY, NOT NULL
payment_id	VARCHAR2(16)	PRIMARY KEY, NOT NULL
payment_reference_num	VARCHAR2(20)	PRIMARY KEY, NOT NULL

4.5 Data Layout

1. employee

employee_id	employee_nric	employee_name	employee_gender	employee_email_address	employee_home_address	employee_age	employee_emergency_contact
2003150001	900101-14-5678	Ahmad Razak	Male	ahmad.razak@mylunas.com.my	Shah Alam, Selangor	36	0129876543
2209140010	940505-02-5566	Nur Aisyah	Female	aisyah@mylunas.com.my	Petaling Jaya, Selangor	30	0198877665
2201200003	930111-03-6677	Lim Wei Jian	Male	weijian@mylunas.com.my.my	Klang, Selangor	31	0126542345
1908120004	880212-10-3344	Siti Hajar	Female	siti.hajar@mylunas.com.my	Subang Jaya, Selangor	38	0166543210
2304030005	920808-07-2211	Mohd Faiz	Male	faiz@mylunas.com.my	Puchong, Selangor	29	0145544332
2111180006	870909-05-4455	Tan Mei Ling	Female	meiling@mylunas.com.my	Cyberjaya, Selangor	33	0165522334
2007060007	910305-08-7788	Ravi Kumar	Male	ravi@mylunas.com.my	Kajang, Selangor	35	0198765432
2139140011	850617-12-9900	Wong Jia Hui	Female	jiahui@mylunas.com.my	Ampang, Selangor	39	0176655443
1806250009	900404-11-1122	Azlan Ismail	Male	azlan@mylunas.com.my	Rawang, Selangor	39	0128877665
2402010008	860223-09-8899	Nurul Syafiqah	Female	syafiqah@mylunas.com.my	Serdang, Selangor	27	0156677889

employee_job_position	employment_status	employee_salary	employee_hire_date	employee_mailing_address	employee_contact_number	department_id	department_code
Project Manager	Active	8500.00	2020-03-15	Shah Alam, Selangor	0112345678	D001	ENG
Accountant	Active	4500.00	2021-05-10	Petaling Jaya, Selangor	0189901243	D004	FIN
Software Engineer	Active	6200.00	2022-01-20	Klang, Selangor	0187766554	D003	IT
Finance Manager	Active	7800.00	2019-08-12	Subang Jaya, Selangor	0198769987	D004	FIN
Site Supervisor	Active	5000.00	2023-04-03	Puchong, Selangor	0139876543	D005	OPS
Procurement Officer	Active	4800.00	2021-11-18	Cyberjaya, Selangor	0147869955	D006	PROC
System Analyst	Active	6500.00	2020-07-06	Kajang, Selangor	0152334554	D003	IT
Finance Manager	Active	3200.00	2024-02-01	Ampang, Selangor	0109765432	D004	FIN
Operations Manager	Active	9200.00	2018-06-25	Rawang, Selangor	0179921006	D005	OPS
Administrative Assistant	Active	5500.00	2022-09-14	Serdang, Selangor	0167778238	D002	HR

2. external_party

party_id	registration_number	company_name	company_email	company_address	contact_person
P001	REG-NUM-1901001	LUNAS Defence Holdings Sdn Bhd	admin@lunasdefence.com.my	Shah Alam, Selangor	Ahmad Razak
P002	REG-NUM-2012002	Malaysian Maritime Enforcement Agency (MMEA)	info@mmea.gov.my	Petaling Jaya, Selangor	Nur Aisyah
P003	REG-NUM-1812003	OceanTech Marine Engineering Sdn Bhd	contact@oceantechmarine.com.my	Klang, Selangor	Lim Wei Jian
P004	REG-NUM-2112004	Boustead Naval Systems Sdn Bhd	enquiry@bousteadnaval.com.my	Subang Jaya, Selangor	Siti Hajar
P005	REG-NUM-1711005	Maritime Shipbuilding Consortium Malaysia	admin@mscmalaysia.com.my	Puchong, Selangor	Mohd Faiz
P006	REG-NUM-2202006	Royal Malaysian Coast Guard	operations@coastguard.gov.my	Cyberjaya, Selangor	Tan Mei Ling
P007	REG-NUM-1908007	Neptune Marine Technology Solutions Sdn Bhd	support@neptunemarine.com.my	Kajang, Selangor	Ravi Kumar
P008	REG-NUM-2006008	Boustead Heavy Industries Corporation (BHIC)	contact@bhic.com.my	Ampang, Selangor	Wong Jia Hui
P009	REG-NUM-1804009	Offshore Engineering & Energy Consortium Malaysia	info@offshoreenergyconsortium.com.my	Rawang, Selangor	Azlan Ismail
P010	REG-NUM-2307010	Malaysian Naval Headquarters	logistics@navy.mil.my	Serdang, Selangor	Nurul Syafiqah

contact_number	financial_status
012-3456789	Good
013-4567890	Good
014-5678901	Average
016-6789012	Excellent
017-7890123	Good
018-8901234	Average
019-9012345	Good
011-1122334	Excellent
012-2233445	Good
013-3344556	Excellent

party_id	registration_ number	company_name	company_email	company_ address	contact_ person
P011	REG-NUM-1911001	NavalTech Marine Engineering Sdn Bhd	contact@navaltechmarine.com.my	Shah Alam, Selangor	Farid Hassan
P012	REG-NUM-2001002	Coastal Defence Supply Agency Malaysia	info@coastaldefence.gov.my	Petaling Jaya, Selangor	Aina Syahirah
P013	REG-NUM-1811003	SteelMarine Fabrication Industries Sdn Bhd	support@steelseamarine.com.my	Klang, Selangor	Lim Zhi Hao
P014	REG-NUM-2110004	Horizon Ship Systems Integration Sdn Bhd	enquiry@horizonshipsystems.com.my	Subang Jaya, Selangor	Nurul Amirah
P015	REG-NUM-1706005	Triton Maritime Engineering Consortium	admin@tritonmaritimeconsortium.com.my	Puchong, Selangor	Kumar Raj
P016	REG-NUM-2205006	Seaguard Equipment Supply Services	sales@seaguardequipment.com.my	Cyberjaya, Selangor	Siti Aisyah
P017	REG-NUM-1910007	RadarWave Technologies Sdn Bhd	contact@radarwavetech.com.my	Kajang, Selangor	Ahmad Syafiq
P018	REG-NUM-2005008	SubSea Defence Systems Manufacturing Sdn Bhd	info@subseadefence.com.my	Ampang, Selangor	Mei Ling Tan
P019	REG-NUM-1811009	Offshore Structures Engineering Alliance Sdn Bhd	support@offshorealliance.com.my	Rawang, Selangor	Ravi Kumar
P020	REG-NUM-2312010	FleetSupport Logistics & Maintenance Sdn Bhd	logistics@fleetsupport.com.my	Serdang, Selangor	Azlan Ismail

contact_number	financial_status
012-1112233	Good
013-2223344	Excellent
014-3334455	Average
016-4445566	Good
017-5556677	Good
018-6667788	Excellent
019-7778899	Average
011-8889900	Good
012-9990011	Good
013-1011122	Excellent

3. Client

client_id	client_reference_num	client_ownership_type	industry_sector	preferred_service_type	last_request_date	project_count
C001	GLC-DEF-4821	Government Linked Company	Defence	Naval Vessel Construction	2026-05-15	3
C002	GOV-MDE-1934	Government Agency	Maritime Defence	Ship Repair & Maintenance	2026-06-01	5
C003	PVT-MRE-7712	Private Limited	Marine Engineering	Hull Fabrication	2026-04-20	2
C004	GLC-DEF-5603	Government Linked Company	Defence	Offshore Patrol Vessel Upgrade	2026-06-10	4
C005	CON-SHB-2291	Consortium	Shipbuilding	Engine Overhaul Services	2026-05-28	6
C006	GOV-CGD-8045	Government Agency	Coast Guard	Vessel Refitting	2026-05-08	3
C007	PVT-MTE-3906	Private Limited	Marine Technology	Radar System Integration	2026-06-12	2
C008	GLC-MDE-1187	Government Linked Company	Maritime Defence	Submarine Support Services	2026-05-18	5
C009	CON-OFE-6673	Consortium	Offshore Engineering	Platform Maintenance	2026-04-30	4
C010	GOV-MSE-9052	Government Agency	Maritime Security	Fleet Maintenance Planning	2026-06-15	7

client_status	client_loyalty	total_outstanding_balance	client_discount_eligibility	total_purchase_value	party_id	registration_number
Active	Platinum	8500000.00	Yes	45000000.00	P001	REG-NUM-1901001
Active	Gold	6200000.00	Yes	32000000.00	P002	REG-NUM-2012002
Active	Silver	1800000.00	Yes	15000000.00	P003	REG-NUM-1812003
Active	Platinum	9200000.00	Yes	50000000.00	P004	REG-NUM-2112004
Active	Gold	7500000.00	Yes	38000000.00	P005	REG-NUM-1711005
Inactive	Silver	2100000.00	Yes	17000000.00	P006	REG-NUM-2202006
Active	Bronze	900000.00	Yes	8000000.00	P007	REG-NUM-1908007
Active	Platinum	9800000.00	Yes	60000000.00	P008	REG-NUM-2006008
Active	Gold	3200000.00	Yes	22000000.00	P009	REG-NUM-1804009
Active	Platinum	11000000.00	Yes	75000000.00	P010	REG-NUM-2307010

4. Supplier

supplier_id	supplier_reference_num	product_category	supply_cycle	last_supply_date	operating_hours	website_url
S001	GLC-ENG-4821	Marine Equipment	Monthly	2026-05-10	0800–1800	www.navaltech1.com
S002	GOV-LOG-1934	Defence Materials	Weekly	2026-06-01	0900–1700	www.maritimegov.my
S003	PVT-STE-7712	Steel & Hull Plates	Monthly	2026-04-20	0800–1700	www.oceanetechmarine.com
S004	GLC-ENG-5603	Ship Engine Systems	Quarterly	2026-06-10	0800–1800	www.bousteadnaval.com
S005	CON-SHP-2291	Ship Components	Monthly	2026-05-28	0900–1800	www.shipconsortium.com
S006	GOV-MAR-8045	Coast Guard Supplies	Weekly	2026-05-08	0900–1700	www.mmea.gov.my
S007	PVT-TEC-3906	Radar & Sensors	Monthly	2026-06-12	0800–1800	www.neptunetech.com
S008	GLC-SUB-1187	Submarine Systems	Quarterly	2026-05-18	0800–1800	www.bhic.com.my
S009	CON-OFF-6673	Offshore Structures	Monthly	2026-04-30	0900–1800	www.offshoreconsortium.com
S010	GOV-FLE-9052	Fleet Maintenance Tools	Weekly	2026-06-15	0900–1700	www.navy.gov.my

supplier_coverage_area	preferred_shipping_method	logistic_partner	price_stability	supplier_type	party_id	registration_ number
Malaysia & SEA	Sea Freight	Pos Malaysia	High	Government Linked Company	P011	REG-NUM-1911001
Malaysia	Land Transport	LHDN Logistics	Medium	Government Agency	P012	REG-NUM-2001002
Malaysia & ASEAN	Sea Freight	GD Express	High	Private Limited	P013	REG-NUM-1811003
Malaysia & Global	Air & Sea Freight	Boustead Logistics	High	Government Linked Company	P014	REG-NUM-2110004
ASEAN Region	Sea Freight	MISC Logistics	Medium	Consortium	P015	REG-NUM-1706005
Malaysia	Land Transport	Royal Logistics Unit	High	Government Agency	P016	REG-NUM-2205006
Asia Pacific	Air Freight	DHL Supply Chain	Medium	Private Limited	P017	REG-NUM-1910007
Global	Sea Freight	BHIC Logistics	High	Government Linked Company	P018	REG-NUM-2005008
ASEAN & Middle East	Sea Freight	Offshore Logistics Alliance	Medium	Consortium	P019	REG-NUM-1811009
Malaysia	Land Transport	Naval Fleet Logistics	High	Government Agency	P020	REG-NUM-2312010

5. department

department_id	department_code	department_ name	department_description	department_type	department_ contact_number	department_official_email
D001	ENG	Engineering	Handles ship design, structural work, and technical engineering	Technical	03-7788 1001	eng@lunas.gov.my
D002	HR	Human Resources	Manages recruitment, staff welfare, and employee relations	Administrative	03-7788 1002	hr@lunas.gov.my
D003	IT	Information Technology	Manages systems, cybersecurity, and internal software	Technical	03-7788 1003	it@lunas.gov.my
D004	FIN	Finance	Handles budgeting, payroll, and financial reporting	Administrative	03-7788 1004	finance@lunas.gov.my
D005	OPS	Operations	Oversees shipyard operations and project execution	Operational	03-7788 1005	ops@lunas.gov.my
D006	PROC	Procurement	Manages supplier contracts and material purchasing	Administrative	03-7788 1006	procurement@lunas.gov.my
D007	QAQC	Quality Assurance	Ensures quality standards in shipbuilding and repair	Technical	03-7788 1007	qaqc@lunas.gov.my
D008	LOG	Logistics	Manages supply chain, delivery, and storage systems	Operational	03-7788 1008	logistics@lunas.gov.my
D009	RND	Research & Development	Develops new naval technologies and innovations	Technical	03-7788 1009	rnd@lunas.gov.my
D010	LEG	Legal & Compliance	Handles contracts, regulations, and legal affairs	Administrative	03-7788 1010	legal@lunas.gov.my

department_office_location	employee_count	department_status
Engineering Block A, LUNAS Administration Complex, Lumut Naval Shipyard	52	ACTIVE
Human Resources Tower, LUNAS Administration Complex, Lumut Naval Shipyard	10	ACTIVE
IT & Systems Building, LUNAS Digital Operations Centre, Lumut Naval Shipyard	25	ACTIVE
Finance Tower, LUNAS Administration Complex, Lumut Naval Shipyard	27	ACTIVE
Operations Control Centre, Main Dockyard Operations Building, Lumut Naval Shipyard	15	SUSPENDED
Procurement & Supply Building, LUNAS Logistics Complex, Lumut Naval Shipyard	13	ACTIVE
Quality Assurance Centre, Shipbuilding Inspection Facility, Lumut Naval Shipyard	10	INACTIVE
Logistics Hub, Central Warehouse Complex, Lumut Naval Shipyard	21	ACTIVE
Research & Development Centre, Innovation & Engineering Park, Lumut Naval Shipyard	43	ACTIVE
Legal & Compliance Office, Corporate Governance Building, Lumut Naval Shipyard	22	INACTIVE

6. vessel

vessel_id	vessel_registration_number	vessel_name	vessel_category	vessel_fuel_type	vessel_fuel_capacity	vessel_capacity	vessel_width	last_maintenance_date
V001	VRN-LUN-001	KD Lunas Defender	Offshore Patrol Vessel	Diesel	120000 L	120 crew	18 m	2026-04-10
V002	VRN-LUN-002	KD Maritime Shield	Frigate	Diesel-Electric	150000 L	180 crew	22 m	2026-03-15
V003	VRN-LUN-003	KD Steel Vanguard	Corvettes	Diesel	90000 L	100 crew	16 m	2026-02-20
V004	VRN-LUN-004	KD Ocean Sentinel	Offshore Support Vessel	Diesel	110000 L	90 crew	20 m	2026-05-01
V005	VRN-LUN-005	KD Naval Guardian	Patrol Vessel	Diesel	80000 L	75 crew	15 m	2026-01-18
V006	VRN-LUN-006	KD Sea Protector	Frigate	Diesel-Electric	140000 L	170 crew	23 m	2026-03-28
V007	VRN-LUN-007	KD Iron Wave	Offshore Patrol Vessel	Diesel	100000 L	110 crew	17 m	2026-04-22
V008	VRN-LUN-008	KD Subsea Hunter	Submarine Support Vessel	Diesel-Electric	160000 L	60 crew	25 m	2026-05-10
V009	VRN-LUN-009	KD Offshore Titan	Offshore Engineering Vessel	Diesel	130000 L	95 crew	19 m	2026-02-05
V010	VRN-LUN-010	KD Fleet Command	Naval Command Vessel	Diesel-Electric	200000 L	200 crew	30 m	2026-06-01

vessel_weight	vessel_distance_max	vessel_status	vessel_build_year	vessel_length	client_id	client_reference_num	contract_id	agreement_reference_number
3500 tonnes	5000 nautical miles	Active	2020	110 m	C001	GLC-DEF-4821	CT001	AGR-DEF-001
4800 tonnes	6500 nautical miles	Active	2019	140 m	C002	GOV-MDE-1934	CT002	AGR-MDE-002
2800 tonnes	4500 nautical miles	Active	2021	95 m	C003	PVT-MRE-7712	CT003	AGR-MRE-003
4000 tonnes	6000 nautical miles	Active	2018	120 m	C004	GLC-DEF-5603	CT004	AGR-DEF-004
2600 tonnes	4000 nautical miles	Maintenance	2022	90 m	C005	CON-SHB-2291	CT005	AGR-SHB-005
5000 tonnes	7000 nautical miles	Active	2017	145 m	C006	GOV-CGD-8045	CT006	AGR-CGD-006
3100 tonnes	4800 nautical miles	Active	2020	105 m	C007	PVT-MTE-3906	CT007	AGR-MTE-007
5200 tonnes	7500 nautical miles	Active	2016	150 m	C008	GLC-MDE-1187	CT008	AGR-MDE-008
3600 tonnes	5500 nautical miles	Active	2019	115 m	C009	CON-OFE-6673	CT009	AGR-OFE-009
6000 tonnes	9000 nautical miles	Active	2015	170 m	C010	GOV-MSE-9052	CT010	AGR-MSE-010

7. service_request

request_id	request_code	request_title	request_service	request_date	request_description	request_priority_level
R001	REQ-DEF-001	Naval Vessel Construction Request	Shipbuilding	2026-05-10	New offshore patrol vessel construction	High
R002	REQ-MDE-002	Ship Repair Contract	Maintenance	2026-05-20	Annual patrol ship repair and servicing	High
R003	REQ-MRE-003	Hull Fabrication Work	Fabrication	2026-04-18	Steel hull fabrication for marine vessel	Medium
R004	REQ-DEF-004	Patrol Vessel Upgrade	Upgrade Services	2026-06-05	Upgrade of offshore patrol vessel systems	High
R005	REQ-SHB-005	Engine Overhaul Project	Engine Services	2026-05-25	Marine engine overhaul works	High
R006	REQ-CGD-006	Coast Guard Refitting	Refitting	2026-05-01	Refitting of coast guard vessels	Medium
R007	REQ-MTE-007	Radar System Integration	Technology	2026-06-10	Integration of radar systems into ships	Medium
R008	REQ-MDE-008	Submarine Support Services	Defence	2026-05-15	Submarine maintenance and support	High
R009	REQ-OFE-009	Offshore Platform Maintenance	Offshore	2026-04-28	Offshore platform servicing	Medium
R010	REQ-MSE-010	Fleet Maintenance Planning	Maritime Security	2026-06-12	Long-term naval fleet maintenance planning	High

estimated_budget	request_approval_status	request_response_date	request_method	client_id	client_reference_num
45000000.00	Approved	2026-05-15	Formal Submission	C001	GLC-DEF-4821
32000000.00	Approved	2026-06-01	Online Portal	C002	GOV-MDE-1934
15000000.00	Approved	2026-04-25	Email Submission	C003	PVT-MRE-7712
50000000.00	Pending	NULL	Formal Submission	C004	GLC-DEF-5603
38000000.00	Approved	2026-05-30	Tender System	C005	CON-SHB-2291
17000000.00	Approved	2026-05-08	Government Portal	C006	GOV-CGD-8045
12000000.00	Approved	2026-06-12	Email Submission	C007	PVT-MTE-3906
60000000.00	Approved	2026-05-18	Formal Submission	C008	GLC-MDE-1187
22000000.00	Approved	2026-04-30	Email Submission	C009	CON-OFE-6673
75000000.00	Approved	2026-06-15	Government Letter	C010	GOV-MSE-9052

8. contract

contract_id	agreement_ reference_ number	contract_title	contract_scope	contract_ sign_date	contract_ end_date	approved_ date
CT001	AGR-DEF-001	Offshore Patrol Vessel Contract	Construction of OPV for defence use	2026-05-16	2028-05-16	2026-05-15
CT002	AGR-MDE-002	Ship Repair Agreement	Maintenance of naval patrol ships	2026-06-02	2027-06-02	2026-06-01
CT003	AGR-MRE-003	Hull Fabrication Contract	Steel hull fabrication for marine vessels	2026-04-26	2027-04-26	2026-04-25
CT004	AGR-DEF-004	Patrol Vessel Upgrade Contract	Upgrade of offshore patrol vessel systems	2026-06-06	2028-06-06	2026-06-05
CT005	AGR-SHB-005	Engine Overhaul Agreement	Marine engine overhaul services	2026-05-30	2027-05-30	2026-05-30
CT006	AGR-CGD-006	Coast Guard Refitting Contract	Refitting of coast guard vessels	2026-05-09	2027-05-09	2026-05-08
CT007	AGR-MTE-007	Radar Integration Contract	Integration of radar systems into naval ships	2026-06-12	2027-06-12	2026-06-12
CT008	AGR-MDE-008	Submarine Support Agreement	Technical support for submarine maintenance	2026-05-18	2028-05-18	2026-05-18
CT009	AGR-OFE-009	Offshore Platform Maintenance Contract	Maintenance of offshore structures	2026-04-30	2027-04-30	2026-04-30
CT010	AGR-MSE-010	Fleet Maintenance Planning Contract	Long-term naval fleet maintenance planning	2026-06-15	2029-06-15	2026-06-15

contract_terms_condition	contract_document_url	contract_status	request_id	request_code	party_id	registration_number
Standard defence procurement terms	www.lunas.gov.my/ct001	Active	R001	REQ-DEF-001	P001	REG-NUM-1901001
Annual maintenance agreement terms	www.lunas.gov.my/ct002	Active	R002	REQ-MDE-002	P002	REG-NUM-2012002
Manufacturing supply conditions	www.lunas.gov.my/ct003	Active	R003	REQ-MRE-003	P003	REG-NUM-1812003
Defence upgrade compliance terms	www.lunas.gov.my/ct004	Active	R004	REQ-DEF-004	P004	REG-NUM-2112004
Engineering service agreement terms	www.lunas.gov.my/ct005	Active	R005	REQ-SHB-005	P005	REG-NUM-1711005
Government refit regulations	www.lunas.gov.my/ct006	Active	R006	REQ-CGD-006	P006	REG-NUM-2202006
Technology integration standards	www.lunas.gov.my/ct007	Active	R007	REQ-MTE-007	P007	REG-NUM-1908007
Submarine maintenance compliance terms	www.lunas.gov.my/ct008	Active	R008	REQ-MDE-008	P008	REG-NUM-2006008
Offshore engineering contract terms	www.lunas.gov.my/ct009	Active	R009	REQ-OFE-009	P009	REG-NUM-1804009
Naval fleet strategic agreement terms	www.lunas.gov.my/ct010	Active	R010	REQ-MSE-010	P010	REG-NUM-2307010

contract_id	agreement_reference_number	contract_title	contract_scope	contract_sign_date	contract_end_date	approved_date
CT011	AGR-DEF-011	Precision Cutting Equipment Contract	Supply of cutting systems and accessories for naval fabrication	2026-05-01	2027-05-01	2026-05-02
CT012	AGR-MDE-012	Marine Engine Diagnostic System Contract	Engine diagnostic tools and testing equipment supply	2026-05-10	2027-05-10	2026-05-11
CT013	AGR-MRE-013	Radar Calibration System Contract	Radar calibration systems for naval operations	2026-04-20	2027-04-20	2026-04-21
CT014	AGR-DEF-014	Hydraulic Press System Contract	Hydraulic press machinery for shipyard fabrication	2026-06-01	2027-06-01	2026-06-01
CT015	AGR-SHB-015	Cable Installation System Contract	Automated cable installation systems for vessels	2026-05-15	2027-05-15	2026-05-16
CT016	AGR-CGD-016	Marine HVAC System Contract	Cooling and HVAC systems for ships	2026-05-03	2027-05-03	2026-05-04
CT017	AGR-MTE-017	Propulsion Balancing System Contract	Propeller balancing equipment and calibration systems	2026-06-05	2027-06-05	2026-06-06
CT018	AGR-MDE-018	Navigation Testing System Contract	Navigation testing and calibration equipment	2026-05-18	2027-05-18	2026-05-19
CT019	AGR-OFE-019	Fuel System Diagnostic Contract	Fuel system diagnostic tools and analyzers	2026-04-25	2027-04-25	2026-04-26
CT020	AGR-MSE-020	Safety Inspection Equipment Contract	Safety inspection and compliance testing equipment	2026-06-10	2027-06-10	2026-06-10

contract_terms_condition	contract_document_url	contract_status	request_id	request_code	party_id	registration_number
Standard procurement terms apply	www.lunas.gov.my/ct011.pdf	Active	-	-	P011	REG-NUM-1911001
Standard procurement terms apply	www.lunas.gov.my/ct012.pdf	Active	-	-	P012	REG-NUM-2001002
Standard procurement terms apply	www.lunas.gov.my/ct013.pdf	Active	-	-	P013	REG-NUM-1811003
Standard procurement terms apply	www.lunas.gov.my/ct014.pdf	Pending	-	-	P014	REG-NUM-2110004
Standard procurement terms apply	www.lunas.gov.my/ct015.pdf	Active	-	-	P015	REG-NUM-1706005
Standard procurement terms apply	www.lunas.gov.my/ct016.pdf	Active	-	-	P016	REG-NUM-2205006
Standard procurement terms apply	www.lunas.gov.my/ct017.pdf	Active	-	-	P017	REG-NUM-1910007
Standard procurement terms apply	www.lunas.gov.my/ct018.pdf	Active	-	-	P018	REG-NUM-2005008
Standard procurement terms apply	www.lunas.gov.my/ct019.pdf	Active	-	-	P019	REG-NUM-1811009
Standard procurement terms apply	www.lunas.gov.my/ct020.pdf	Pending	-	-	P020	REG-NUM-2312010

9. project

project_id	project_reference_num	project_name	project_description	project_type	project_status	project_start_date	client_id	client_reference_num
PR001	PROJ-DEF-001	OPV Construction Project	Build offshore patrol vessel for defence use	Shipbuilding	In Progress	2026-05-16	C001	GLC-DEF-4821
PR002	PROJ-MDE-002	Naval Ship Repair Project	Repair and maintenance of naval patrol ships	Maintenance	In Progress	2026-06-02	C002	GOV-MDE-1934
PR003	PROJ-MRE-003	Hull Fabrication Project	Fabrication of steel hull structures	Manufacturing	Completed	2026-04-26	C003	PVT-MRE-7712
PR004	PROJ-DEF-004	Patrol Vessel Upgrade Project	Upgrade systems of offshore patrol vessel	Upgrade	In Progress	2026-06-06	C004	GLC-DEF-5603
PR005	PROJ-SHB-005	Engine Overhaul Project	Overhaul marine engines for naval vessels	Maintenance	Completed	2026-05-30	C005	CON-SHB-2291
PR006	PROJ-CGD-006	Coast Guard Refitting Project	Refitting of coast guard vessels	Refitting	Completed	2026-05-09	C006	GOV-CGD-8045
PR007	PROJ-MTE-007	Radar Integration Project	Integrate radar systems into naval ships	Technology	In Progress	2026-06-12	C007	PVT-MTE-3906
PR008	PROJ-MDE-008	Submarine Support Project	Submarine maintenance and technical support	Defence	In Progress	2026-05-18	C008	GLC-MDE-1187
PR009	PROJ-OFE-009	Offshore Maintenance Project	Maintenance of offshore structures	Offshore	Completed	2026-04-30	C009	CON-OFE-6673
PR010	PROJ-MSE-010	Fleet Maintenance Planning Project	Long-term naval fleet maintenance strategy	Strategic	Planned	2026-06-15	C010	GOV-MSE-9052

project_due_date	project_finished_date	project_priority	project_budget	contract_id	agreement_reference_number	request_id	request_code
2028-05-16	2028-05-16	High	45000000.00	CT001	AGR-DEF-001	R001	REQ-DEF-001
2027-06-02	NULL	High	32000000.00	CT002	AGR-MDE-002	R002	REQ-MDE-002
2027-04-26	2027-04-26	Medium	15000000.00	CT003	AGR-MRE-003	R003	REQ-MRE-003
2028-06-06	NULL	High	50000000.00	CT004	AGR-DEF-004	R004	REQ-DEF-004
2027-05-30	2027-05-30	High	38000000.00	CT005	AGR-SHB-005	R005	REQ-SHB-005
2027-05-09	2027-05-09	Medium	17000000.00	CT006	AGR-CGD-006	R006	REQ-CGD-006
2027-06-12	NULL	Medium	12000000.00	CT007	AGR-MTE-007	R007	REQ-MTE-007
2028-05-18	NULL	High	60000000.00	CT008	AGR-MDE-008	R008	REQ-MDE-008
2027-04-30	2027-04-30	Medium	22000000.00	CT009	AGR-OFE-009	R009	REQ-OFE-009
2029-06-15	NULL	High	75000000.00	CT010	AGR-MSE-010	R010	REQ-MSE-010

10. work_order

work_order_id	work_order_num	work_order_title	work_order_details	work_order_type	work_order_status	work_priority_level	work_start_date
WO001	WON-DEF-001	Vessel Construction Work Order	Construction of offshore patrol vessel hull and systems	Construction	In Progress	High	2026-05-16
WO002	WON-MDE-002	Ship Repair Work Order	Repair and maintenance of naval patrol ship systems	Maintenance	Completed	High	2026-06-02
WO003	WON-MRE-003	Hull Fabrication Work Order	Fabrication of steel hull sections	Fabrication	Completed	Medium	2026-04-26
WO004	WON-DEF-004	Patrol Vessel Upgrade Work Order	Upgrade radar and navigation systems	Upgrade	In Progress	High	2026-06-06
WO005	WON-SHB-005	Engine Overhaul Work Order	Overhaul marine propulsion engines	Maintenance	Completed	High	2026-05-30
WO006	WON-CGD-006	Coast Guard Refitting Work Order	Refitting and system replacement	Refitting	Completed	Medium	2026-05-09
WO007	WON-MTE-007	Radar Integration Work Order	Install radar systems on vessels	Technology	In Progress	Medium	2026-06-12
WO008	WON-MDE-008	Submarine Support Work Order	Submarine maintenance and support operations	Defence	In Progress	High	2026-05-18
WO009	WON-OFE-009	Offshore Maintenance Work Order	Offshore platform structural maintenance	Offshore	Completed	Medium	2026-04-30
WO010	WON-MSE-010	Fleet Maintenance Planning Work Order	Long-term naval fleet maintenance planning	Strategic	Planned	High	2026-06-15

work_due_date	work_actual_ end_date	vessel_id	vessel_registration_number	project_id	project_ reference_num	contract_id	agreement_ reference_ number
2028-05-16	NULL	V001	VRN-LUN-001	PR001	PROJ-DEF-001	CT001	AGR-DEF-001
2027-06-02	2027-06-02	V002	VRN-LUN-002	PR002	PROJ-MDE-002	CT002	AGR-MDE-002
2027-04-26	2027-04-26	V003	VRN-LUN-003	PR003	PROJ-MRE-003	CT003	AGR-MRE-003
2028-06-06	NULL	V004	VRN-LUN-004	PR004	PROJ-DEF-004	CT004	AGR-DEF-004
2027-05-30	2027-05-30	V005	VRN-LUN-005	PR005	PROJ-SHB-005	CT005	AGR-SHB-005
2027-05-09	2027-05-09	V006	VRN-LUN-006	PR006	PROJ-CGD-006	CT006	AGR-CGD-006
2027-06-12	NULL	V007	VRN-LUN-007	PR007	PROJ-MTE-007	CT007	AGR-MTE-007
2028-05-18	NULL	V008	VRN-LUN-008	PR008	PROJ-MDE-008	CT008	AGR-MDE-008
2027-04-30	2027-04-30	V009	VRN-LUN-009	PR009	PROJ-OFE-009	CT009	AGR-OFE-009
2029-06-15	NULL	V010	VRN-LUN-010	PR010	PROJ-MSE-010	CT010	AGR-MSE-010

11. work_report

report_id	report_reference_num	report_type	report_title	report_ created_date	report_description	report_ updated_date
RPT001	REP-MNT-001	Maintenance	Engine Performance Report	2026-05-20	Report on vessel engine efficiency and issues	2026-05-22
RPT002	REP-INS-002	Inspection	Hull Integrity Report	2026-06-01	Structural inspection of ship hull condition	2026-06-02
RPT003	REP-PRG-003	Progress	Project Progress Report	2026-04-28	Progress update on fabrication works	2026-04-30
RPT004	REP-HSE-004	Safety	Safety Compliance Report	2026-06-10	Safety audit for shipyard operations	2026-06-11
RPT005	REP-TEC-005	Technical	Radar System Report	2026-05-15	Evaluation of radar integration performance	2026-05-16
RPT006	REP-MNT-006	Maintenance	Submarine Support Report	2026-05-18	Submarine maintenance findings	2026-05-19
RPT007	REP-OPR-007	Operational	Fleet Readiness Report	2026-06-12	Assessment of fleet operational readiness	2026-06-13
RPT008	REP-ENG-008	Engineering	Offshore Structure Report	2026-04-30	Structural analysis of offshore platform	2026-05-01
RPT009	REP-LOG-009	Logistics	Supply Chain Report	2026-05-25	Review of logistics and supply efficiency	2026-05-26
RPT010	REP-STR-010	Strategic	Naval Fleet Strategy Report	2026-06-15	Long-term fleet planning analysis	2026-06-16

report_attachment_reference_url	suggestion_action	report_status	vessel_id	vessel_registration_number	work_order_id	work_order_num
www.lunas.gov.my/rpt001.pdf	Repair engine components	Completed	V001	VRN-LUN-001	WO001	WON-DEF-001
www.lunas.gov.my/rpt002.pdf	Reinforce hull structure	Completed	V002	VRN-LUN-002	WO002	WON-MDE-002
www.lunas.gov.my/rpt003.pdf	Continue fabrication schedule	Completed	V003	VRN-LUN-003	WO003	WON-MRE-003
www.lunas.gov.my/rpt004.pdf	Improve safety protocols	Completed	V004	VRN-LUN-004	WO004	WON-DEF-004
www.lunas.gov.my/rpt005.pdf	Optimize radar calibration	Completed	V005	VRN-LUN-005	WO005	WON-SHB-005
www.lunas.gov.my/rpt006.pdf	Schedule submarine servicing	Completed	V006	VRN-LUN-006	WO006	WON-CGD-006
www.lunas.gov.my/rpt007.pdf	Increase fleet readiness checks	Completed	V007	VRN-LUN-007	WO007	WON-MTE-007
www.lunas.gov.my/rpt008.pdf	Strengthen offshore structures	Completed	V008	VRN-LUN-008	WO008	WON-MDE-008
www.lunas.gov.my/rpt009.pdf	Improve supply chain efficiency	Completed	V009	VRN-LUN-009	WO009	WON-OFE-009
www.lunas.gov.my/rpt010.pdf	Revise fleet strategy planning	Completed	V010	VRN-LUN-010	WO010	WON-MSE-010

project_id	project_reference_num	invoice_id	payment_reference_num	Invoice_number	agreement_reference_number
PR001	PROJ-DEF-001	INV001	PAY-COM-001	INV-LUN-001	AGR-DEF-001
PR002	PROJ-MDE-002	INV002	PAY-PAR-002	INV-LUN-002	AGR-MDE-002
PR003	PROJ-MRE-003	INV003	PAY-COM-003	INV-LUN-003	AGR-MRE-003
PR004	PROJ-DEF-004	INV004	PAY-PAR-004	INV-LUN-004	AGR-DEF-004
PR005	PROJ-SHB-005	INV005	PAY-COM-005	INV-LUN-005	AGR-SHB-005
PR006	PROJ-CGD-006	INV006	PAY-COM-006	INV-LUN-006	AGR-CGD-006
PR007	PROJ-MTE-007	INV007	PAY-COM-007	INV-LUN-007	AGR-MTE-007
PR008	PROJ-MDE-008	INV008	PAY-PAR-008	INV-LUN-008	AGR-MDE-008
PR009	PROJ-OFE-009	INV009	PAY-COM-009	INV-LUN-009	AGR-OFE-009
PR010	PROJ-MSE-010	INV010	PAY-PAR-010	INV-LUN-010	AGR-MSE-010

employee_id	employee_nric	client_id	client_reference_num	contract_id	payment_id
2003150001	900101-14-5678	C001	GLC-DEF-4821	CT001	PAY001
1908120004	880212-10-3344	C002	GOV-MDE-1934	CT002	PAY002
2007060007	910305-08-7788	C003	PVT-MRE-7712	CT003	PAY003
2139140011	850617-12-9900	C004	GLC-DEF-5603	CT004	PAY004
2304030005	920808-07-2211	C005	CON-SHB-2291	CT005	PAY005
2111180006	870909-05-4455	C006	GOV-CGD-8045	CT006	PAY006
2201200003	930111-03-6677	C007	PVT-MTE-3906	CT007	PAY007
2402010008	860223-09-8899	C008	GLC-MDE-1187	CT008	PAY008
1806250009	900404-11-1122	C009	CON-OFE-6673	CT009	PAY009
2209140010	940505-02-5566	C010	GOV-MSE-9052	CT010	PAY010

12. invoice

invoice_id	invoice_number	invoice_title	subtotal_amount	tax_amount	discount_amount	invoice_date	outstanding_balance
INV001	INV-LUN-001	Vessel Construction Invoice	45000000.00	2700000.00	500000.00	2026-05-20	0.00
INV002	INV-LUN-002	Ship Repair Invoice	32000000.00	1920000.00	300000.00	2026-06-02	500000.00
INV003	INV-LUN-003	Hull Fabrication Invoice	15000000.00	900000.00	150000.00	2026-04-26	0.00
INV004	INV-LUN-004	Patrol Vessel Upgrade Invoice	50000000.00	3000000.00	750000.00	2026-06-06	1000000.00
INV005	INV-LUN-005	Engine Overhaul Invoice	38000000.00	2280000.00	400000.00	2026-05-30	0.00
INV006	INV-LUN-006	Coast Guard Refitting Invoice	17000000.00	1020000.00	200000.00	2026-05-09	0.00
INV007	INV-LUN-007	Radar Integration Invoice	12000000.00	720000.00	100000.00	2026-06-12	0.00
INV008	INV-LUN-008	Submarine Support Invoice	60000000.00	3600000.00	800000.00	2026-05-18	1200000.00
INV009	INV-LUN-009	Offshore Maintenance Invoice	22000000.00	1320000.00	250000.00	2026-04-30	0.00
INV010	INV-LUN-010	Fleet Strategy Invoice	75000000.00	4500000.00	1000000.00	2026-06-15	1500000.00

total_amount	invoice_description	invoice_status	invoice_due_date	invoice_type	project_id
46270000.00	Payment for vessel construction project	Paid	2026-06-20	Sales	PR001
33620000.00	Maintenance and repair services	Partial	2026-07-02	Sales	PR002
15785000.00	Hull fabrication materials and labor	Paid	2026-05-26	Sales	PR003
52250000.00	Upgrade of naval vessel systems	Partial	2026-07-06	Sales	PR004
39880000.00	Engine overhaul and servicing	Paid	2026-06-30	Sales	PR005
17730000.00	Coast guard vessel refitting	Paid	2026-06-09	Sales	PR006
12720000.00	Radar system integration works	Paid	2026-07-12	Sales	PR007
60720000.00	Submarine support operations	Partial	2026-06-18	Sales	PR008
23070000.00	Offshore maintenance services	Paid	2026-05-30	Sales	PR009
78500000.00	Long-term fleet strategy implementation	Partial	2026-07-15	Sales	PR010

project_reference_num	contract_id	agreement_reference_number	party_id	registration_number	purchase_id	purchase_reference_num
PROJ-DEF-001	CT001	AGR-DEF-001	P001	REG-NUM-1901001	-	-
PROJ-MDE-002	CT002	AGR-MDE-002	P002	REG-NUM-2012002	-	-
PROJ-MRE-003	CT003	AGR-MRE-003	P003	REG-NUM-1812003	-	-
PROJ-DEF-004	CT004	AGR-DEF-004	P004	REG-NUM-2112004	-	-
PROJ-SHB-005	CT005	AGR-SHB-005	P005	REG-NUM-1711005	-	-
PROJ-CGD-006	CT006	AGR-CGD-006	P006	REG-NUM-2202006	-	-
PROJ-MTE-007	CT007	AGR-MTE-007	P007	REG-NUM-1908007	-	-
PROJ-MDE-008	CT008	AGR-MDE-008	P008	REG-NUM-2006008	-	-
PROJ-OFE-009	CT009	AGR-OFE-009	P009	REG-NUM-1804009	-	-
PROJ-MSE-010	CT010	AGR-MSE-010	P010	REG-NUM-2307010	-	-

invoice_id	invoice_number	invoice_title	subtotal_amount	tax_amount	discount_amount	invoice_date	outstanding_balance
INV011	INV-PO-001	Precision Cutting Equipment Invoice	1200000.00	72000.00	60000.00	2026-05-01	1222000.00
INV012	INV-PO-002	Marine Engine Diagnostic Invoice	950000.00	57000.00	28500.00	2026-05-10	977000.00
INV013	INV-PO-003	Radar Calibration System Invoice	1500000.00	90000.00	60000.00	2026-04-20	1515000.00
INV014	INV-PO-004	Hydraulic Press System Invoice	1800000.00	108000.00	36000.00	2026-06-01	1848000.00
INV015	INV-PO-005	Cable Installation System Invoice	800000.00	48000.00	24000.00	2026-05-15	823000.00
INV016	INV-PO-006	Marine HVAC System Invoice	2100000.00	126000.00	42000.00	2026-05-03	2126000.00
INV017	INV-PO-007	Propulsion Balancing System Invoice	1100000.00	66000.00	44000.00	2026-06-05	1126000.00
INV018	INV-PO-008	Navigation Testing System Invoice	2500000.00	150000.00	125000.00	2026-05-18	2530000.00
INV019	INV-PO-009	Fuel Diagnostic Equipment Invoice	900000.00	54000.00	27000.00	2026-04-25	934000.00
INV020	INV-PO-010	Safety Inspection Equipment Invoice	3000000.00	180000.00	150000.00	2026-06-10	3030000.00

total_amount	invoice_description	invoice _status	invoice_due_date	Invoice_type	project_id	project _reference_ num
1212000.00	Cutting equipment supply for shipyard	Paid	2026-06-01	Purchase	PR001	PROJ-DEF-001
978500.00	Engine diagnostic tools procurement	Paid	2026-06-10	Purchase	PR002	PROJ-MDE-002
1530000.00	Radar calibration systems supply	Paid	2026-05-20	Purchase	PR003	PROJ-MRE-003
1902000.00	Hydraulic press system procurement	Pending	2026-07-01	Purchase	PR004	PROJ-DEF-004
824000.00	Cable installation system purchase	Paid	2026-06-15	Purchase	PR005	PROJ-SHB-005
2184000.00	HVAC system procurement	Paid	2026-06-03	Purchase	PR006	PROJ-CGD-006
1122000.00	Propulsion balancing system purchase	Paid	2026-07-05	Purchase	PR007	PROJ-MTE-007
2625000.00	Navigation testing equipment supply	Paid	2026-06-18	Purchase	PR008	PROJ-MDE-008
927000.00	Fuel system diagnostic tools	Paid	2026-05-25	Purchase	PR009	PROJ-OFE-009
3030000.00	Safety inspection equipment procurement	Pending	2026-07-10	Purchase	PR010	PROJ-MSE-010

contract _id	agreement_ reference_number	party_id	registration_ number	purchase_ id	purchase_ reference_num
CT011	AGR-DEF-011	P011	REG-NUM-1911001	PO001	PO-EQP-001
CT012	AGR-MDE-012	P012	REG-NUM-2001002	PO002	PO-EQP-002
CT013	AGR-MRE-013	P013	REG-NUM-1811003	PO003	PO-EQP-003
CT014	AGR-DEF-014	P014	REG-NUM-2110004	PO004	PO-EQP-004
CT015	AGR-SHB-015	P015	REG-NUM-1706005	PO005	PO-EQP-005
CT016	AGR-CGD-016	P016	REG-NUM-2205006	PO006	PO-EQP-006
CT017	AGR-MTE-017	P017	REG-NUM-1910007	PO007	PO-EQP-007
CT018	AGR-MDE-018	P018	REG-NUM-2005008	PO008	PO-EQP-008
CT019	AGR-OFE-019	P019	REG-NUM-1811009	PO009	PO-EQP-009
CT020	AGR-MSE-020	P020	REG-NUM-2312010	PO010	PO-EQP-010

13. payment

payment_id	payment_reference_num	payment_date	payment_timestamp	payment_method	payment_status
PAY001	PAY-COM-001	2026-05-21	2026-05-21 10:15:00	Bank Transfer	Completed
PAY002	PAY-PAR-002	2026-06-03	2026-06-03 14:20:00	Bank Transfer	Partial
PAY003	PAY-COM-003	2026-04-27	2026-04-27 09:45:00	Online Banking	Completed
PAY004	PAY-PAR-004	2026-06-07	2026-06-07 16:10:00	Bank Transfer	Partial
PAY005	PAY-COM-005	2026-05-31	2026-05-31 11:00:00	Cheque	Completed
PAY006	PAY-COM-006	2026-05-10	2026-05-10 13:30:00	Bank Transfer	Completed
PAY007	PAY-COM-007	2026-06-13	2026-06-13 15:25:00	Online Banking	Completed
PAY008	PAY-PAR-008	2026-05-19	2026-05-19 10:50:00	Bank Transfer	Partial
PAY009	PAY-COM-009	2026-05-01	2026-05-01 12:05:00	Online Banking	Completed
PAY010	PAY-PAR-010	2026-06-16	2026-06-16 09:00:00	Bank Transfer	Partial

payment_reference	transaction_type	payment_currency_code	payment_amount	party_id	registration_number
REF-INV-001	Credit	MYR	46270000.00	P001	REG-NUM-1901001
REF-INV-002	Credit	MYR	33120000.00	P002	REG-NUM-2012002
REF-INV-003	Credit	MYR	15785000.00	P003	REG-NUM-1812003
REF-INV-004	Credit	MYR	51250000.00	P004	REG-NUM-2112004
REF-INV-005	Credit	MYR	39880000.00	P005	REG-NUM-1711005
REF-INV-006	Credit	MYR	17730000.00	P006	REG-NUM-2202006
REF-INV-007	Credit	MYR	12720000.00	P007	REG-NUM-1908007
REF-INV-008	Credit	MYR	59520000.00	P008	REG-NUM-2006008
REF-INV-009	Credit	MYR	23070000.00	P009	REG-NUM-1804009
REF-INV-010	Credit	MYR	77000000.00	P010	REG-NUM-2307010

14. purchase_order

purchase_id	purchase_reference_num	purchase_order_name	order_details	agreed_tax_rate	agreed_service_fee	agreed_discount_percentage	order_fulfillment_status
PO001	PO-EQP-001	Precision Cutting Equipment Order	Armoured steel cutter and cutting accessories	6%	2.5%	5%	Fulfilled
PO002	PO-EQP-002	Marine Engine Diagnostic System Order	Marine engine test bench and diagnostic modules	6%	2.0%	3%	Fulfilled
PO003	PO-EQP-003	Radar Calibration System Order	Naval radar calibration equipment and software	6%	2.5%	4%	Fulfilled
PO004	PO-EQP-004	Hydraulic Press System Order	Ship hull hydraulic press and control components	6%	3.0%	2%	Pending
PO005	PO-EQP-005	Automated Cable Installation System Order	Automated cable laying system and installation tools	6%	2.0%	3%	Fulfilled
PO006	PO-EQP-006	Marine HVAC & Cooling System Order	Ship cooling and HVAC units with spare parts	6%	2.0%	2%	Fulfilled
PO007	PO-EQP-007	Propulsion Balancing System Order	Propeller dynamic balancing equipment and sensors	6%	3.0%	4%	Fulfilled
PO008	PO-EQP-008	Navigation Testing System Order	Navigation system testing suite and calibration devices	6%	2.5%	5%	Fulfilled
PO009	PO-EQP-009	Fuel System Diagnostic Equipment Order	Fuel injection diagnostic analyzer and testing kits	6%	2.0%	3%	Fulfilled
PO010	PO-EQP-010	Safety Inspection Equipment Order	Safety inspection kits and compliance testing tools	6%	3.5%	6%	Pending

order_date	order_approval_status	order_ approval_date	supplier _id	supplier_ reference_num	project_id	Project_reference _num	contract_id	agreement_reference_number
2026-05-01	APPROVED	2026-05-12	S001	GLC-ENG-4821	PR001	PROJ-DEF-001	CT011	AGR-DEF-011
2026-05-10	APPROVED	2026-05-11	S002	GOV-LOG-1934	PR002	PROJ-MDE-002	CT012	AGR-MDE-012
2026-04-20	APPROVED	2026-04-21	S003	PVT-STE-7712	PR003	PROJ-MRE-003	CT013	AGR-MRE-013
2026-06-01	REJECTED	NULL	S004	GLC-ENG-5603	PR004	PROJ-DEF-004	CT014	AGR-DEF-014
2026-05-15	APPROVED	2026-05-16	S005	CON-SHP-2291	PR005	PROJ-SHB-005	CT015	AGR-SHB-015
2026-05-03	APPROVED	2026-05-04	S006	GOV-MAR-8045	PR006	PROJ-CGD-006	CT016	AGR-CGD-016
2026-06-05	APPROVED	2026-06-06	S007	PVT-TEC-3906	PR007	PROJ-MTE-007	CT017	AGR-MTE-017
2026-05-18	APPROVED	2026-05-19	S008	GLC-SUB-1187	PR008	PROJ-MDE-008	CT018	AGR-MDE-018
2026-04-25	APPROVED	2026-04-26	S009	CON-OFF-6673	PR009	PROJ-OFE-009	CT019	AGR-OFE-019
2026-06-10	REJECTED	NULL	S010	GOV-FLE-9052	PR010	PROJ-MSE-010	CT020	AGR-MSE-020

15. inventory

inventory_item_id	inventory_reference_number	item_name	storage_location	unit_cost	item_description
INV001	INV-MAT-001	Steel Plate A36	Warehouse A	2500.00	Marine-grade steel plate for hull construction
INV002	INV-MAT-002	Engine Gear Set	Warehouse B	18500.00	Main propulsion engine gear components
INV003	INV-MAT-003	Radar Module X1	Warehouse C	45000.00	Naval radar system module
INV004	INV-MAT-004	Hydraulic Pump	Warehouse B	8200.00	Hydraulic system pump for vessel operations
INV005	INV-MAT-005	Marine Cable Set	Warehouse A	1200.00	Electrical wiring for ship systems
INV006	INV-MAT-006	Cooling System Unit	Warehouse C	15000.00	Engine cooling system component
INV007	INV-MAT-007	Propeller Blade	Warehouse D	22000.00	High-strength alloy propeller blade
INV008	INV-MAT-008	Navigation Sensor	Warehouse C	38000.00	GPS and navigation sensor unit
INV009	INV-MAT-009	Fuel Pump Assembly	Warehouse B	9800.00	Fuel distribution system component
INV010	INV-MAT-010	Safety Helmet Set	Warehouse A	150.00	Personal protective equipment

inventory_item_id	Inventory _reference_number	item_name	storage_location	unit_cost	item_description
INV011	INV-EQP-011	Armoured Steel Cutter	Fabrication Bay A	32000.00	Heavy-duty steel cutting equipment for hull construction
INV012	INV-EQP-012	Marine Engine Test Bench	Engine Workshop B	88000.00	Engine diagnostics and testing system
INV013	INV-EQP-013	Naval Radar Calibration Unit	Radar Lab C	120000.00	Radar system calibration and testing unit
INV014	INV-EQP-014	Ship Hull Hydraulic Press	Structural Bay A	95000.00	Hydraulic press for hull shaping and assembly
INV015	INV-EQP-015	Automated Cable Laying System	Electrical Yard B	54000.00	Automated system for ship cable installation
INV016	INV-EQP-016	Ship Cooling & HVAC Unit	HVAC Station C	61000.00	Cooling and ventilation system for vessels
INV017	INV-EQP-017	Propeller Dynamic Balancer	Propulsion Lab D	77000.00	Balancing system for marine propellers
INV018	INV-EQP-018	Navigation System Testing Suite	Navigation Center C	110000.00	Testing suite for marine navigation systems
INV019	INV-EQP-019	Fuel Injection Diagnostic Analyzer	Engine Diagnostics B	45000.00	Fuel system testing and diagnostics equipment
INV020	INV-EQP-020	Safety & Inspection Kit	Safety Unit A	8000.00	PPE and safety inspection tools for shipyard use

16. Material

material_item_id	material_code	material_item_type	material_usage_purpose	material_unit_measurement	quantity_available	quantity_reserved	material_current_condition	material_hazard_category	shelf_life_days
MAT001	STL-A36	Metal	Hull Construction	Tonnes	500	50	Good	Low	3650
MAT002	ENG-GR	Mechanical	Engine Assembly	Units	120	10	Good	Medium	3000
MAT003	RAD-MOD	Electronic	Surveillance System	Units	80	5	Excellent	High	2000
MAT004	HYD-PMP	Mechanical	Hydraulic System	Units	60	8	Good	Medium	2500
MAT005	CAB-MAR	Electrical	Ship Wiring	Rolls	300	20	Fair	Low	1500
MAT006	CLG-UNIT	Mechanical	Cooling System	Units	70	6	Good	Medium	2700
MAT007	PRP-BLD	Mechanical	Propulsion	Units	40	4	Excellent	High	3200
MAT008	NAV-SEN	Electronic	Navigation System	Units	90	7	Good	Low	4000
MAT009	FUL-PMP	Mechanical	Fuel System	Units	110	12	Good	Medium	2800
MAT010	SAF-HLM	Safety	Personnel Protection	Sets	500	60	Excellent	Low	5000

storage_condition	expiry_date	equipment_id		equipment_serial_num	inventory_item_id	inventory_reference_number
Dry Warehouse	2035-01-01	NULL		NULL	INV001	INV-MAT-001
Controlled Temperature	2034-06-01	NULL	NULL	INV002		INV-MAT-002
Secure Vault	2032-12-01	NULL	NULL	INV003		INV-MAT-003
Dry Warehouse	2033-08-01	NULL	NULL	INV004		INV-MAT-004
Dry Storage	2030-01-01	NULL	NULL	INV005		INV-MAT-005
Cooling Facility	2033-11-01	NULL	NULL	INV006		INV-MAT-006
Reinforced Storage	2036-01-01	NULL	NULL	INV007		INV-MAT-007
Secure Warehouse	2037-05-01	NULL	NULL	INV008		INV-MAT-008
Dry Warehouse	2034-02-01	NULL	NULL	INV009		INV-MAT-009
PPE Storage Area	2038-01-01	NULL	NULL	INV010		INV-MAT-010

17. Equipment

equipment_id	equipment_serial_num	equipment_type	manufacturer	purchase_date	warranty_period	warranty_expiry_date	last_maintenance_date	equipment_condition	delivery_id	tracking_number
EQ001	EQ-LUN-001-STEEL	Precision Cutting Equipment	Bosch	2021-03-10	3 Years	2024-03-10	2026-04-10	Good	DEL001	TRK-LUN-001
EQ002	EQ-LUN-002-ENG	Marine Engine Diagnostic System	Siemens	2020-06-15	5 Years	2025-06-15	2026-03-15	Excellent	DEL002	TRK-LUN-002
EQ003	EQ-LUN-003-RAD	Radar Calibration System	Thales	2022-01-20	4 Years	2026-01-20	2026-02-20	Good	DEL003	TRK-LUN-003
EQ004	EQ-LUN-004-HYD	Hydraulic Press System	Caterpillar	2019-11-05	5 Years	2024-11-05	2026-05-01	Fair	DEL004	TRK-LUN-004
EQ005	EQ-LUN-005-CAB	Automated Cable Installation System	Schneider Electric	2023-02-12	3 Years	2026-02-12	2026-01-18	Good	DEL005	TRK-LUN-005
EQ006	EQ-LUN-006-CLG	Marine HVAC & Cooling System	Daikin	2021-08-18	4 Years	2025-08-18	2026-03-28	Excellent	DEL006	TRK-LUN-006
EQ007	EQ-LUN-007-PRP	Propulsion Balancing System	Rolls-Royce Marine	2018-07-22	6 Years	2024-07-22	2026-04-22	Good	DEL007	TRK-LUN-007
EQ008	EQ-LUN-008-NAV	Navigation Testing System	Garmin	2022-09-30	3 Years	2025-09-30	2026-05-10	Excellent	DEL008	TRK-LUN-008
EQ009	EQ-LUN-009-FUL	Fuel System Diagnostic Equipment	Delphi	2020-12-01	5 Years	2025-12-01	2026-02-05	Good	DEL009	TRK-LUN-009
EQ010	EQ-LUN-010-SAF	Safety Inspection Equipment	3M Safety	2024-01-10	3 Years	2027-01-10	2026-06-01	Excellent	DEL010	TRK-LUN-010

equipment_status	equipment_used_location	model_number	expected_lifespan	supplier_id	supplier_reference_num	purchase_id	purchase_reference_num	invoice_id	invoice_number	inventory_item_id	inventory_reference_number
Active	Warehouse B	BOS-MTE-100	10 Years	S001	SUP-DEF-001	PO001	PO-EQP-001	INV011	INV-PO-001	INV011	INV-EQP-011
Active	Warehouse A	SI-MNT-200	12 Years	S002	SUP-MNT-002	PO002	PO-EQP-002	INV012	INV-PO-002	INV022	INV-EQP-012
Active	Dry Dock B	TH-MRE-300	10 Years	S003	SUP-MRE-003	PO003	PO-EQP-003	INV013	INV-PO-003	INV033	INV-EQP-013
Maintenance	Workshop	CAT-DEF-400	8 Years	S004	SUP-DEF-004	PO004	PO-EQP-004	INV014	INV-PO-004	INV044	INV-EQP-014
Active	Workshop	SE-SHB-500	10 Years	S005	SUP-SHB-005	PO005	PO-EQP-005	INV015	INV-PO-005	INV055	INV-EQP-015
Active	Repair Bay	DAI-CGD-600	11 Years	S006	SUP-CGD-006	PO006	PO-EQP-006	INV016	INV-PO-006	INV066	INV-EQP-016
Maintenance	Workshop	RR-MTE-700	9 Years	S007	SUP-MTE-007	PO007	PO-EQP-007	INV017	INV-PO-007	INV077	INV-EQP-017
Active	Project Site A	GAR-MDE-800	10 Years	S008	SUP-MDE-008	PO008	PO-EQP-008	INV018	INV-PO-008	INV088	INV-EQP-018
Active	Service Bay	DEL-OFE-900	10 Years	S009	SUP-OFE-009	PO009	PO-EQP-009	INV019	INV-PO-009	INV099	INV-EQP-019
Active	Inspection Area	3M-MSE-1000	8 Years	S010	SUP-MSE-010	PO010	PO-EQP-010	INV020	INV-PO-010	INV020	INV-EQP-020

18. delivery

delivery_id	tracking_number	delivery_shipping_date	delivery_address	delivery_method	delivery_status	estimated_arrival_date	actual_arrived_date	delivery_fee	delivery_contact_number
DEL001	TRK-LUN-001	2026-05-02	LUNAS Shipyard, Lumut, Perak	Freight	Delivered	2026-05-05	2026-05-05	2500.00	03-77881001
DEL002	TRK-LUN-002	2026-05-11	LUNAS Shipyard, Lumut, Perak	Air Freight	Delivered	2026-05-14	2026-05-14	3200.00	03-77881002
DEL003	TRK-LUN-003	2026-04-21	LUNAS Warehouse, Lumut	Sea Freight	Delivered	2026-04-25	2026-04-25	1800.00	03-77881003
DEL004	TRK-LUN-004	2026-06-02	LUNAS Shipyard, Lumut, Perak	Truck	In Transit	2026-06-06	NULL	4000.00	03-77881004
DEL005	TRK-LUN-005	2026-05-16	LUNAS Storage Facility	Freight	Delivered	2026-05-19	2026-05-19	2200.00	03-77881005
DEL006	TRK-LUN-006	2026-05-04	LUNAS Shipyard, Lumut	Sea Freight	Delivered	2026-05-08	2026-05-08	2600.00	03-77881006
DEL007	TRK-LUN-007	2026-06-06	LUNAS Warehouse, Lumut	Air Freight	Delivered	2026-06-09	2026-06-09	3000.00	03-77881007
DEL008	TRK-LUN-008	2026-05-19	LUNAS Offshore Yard	Sea Freight	Delivered	2026-05-23	2026-05-23	3500.00	03-77881008
DEL009	TRK-LUN-009	2026-04-26	LUNAS Storage Hub	Truck	Delivered	2026-04-30	2026-04-30	2000.00	03-77881009
DEL010	TRK-LUN-010	2026-06-11	LUNAS Shipyard HQ	Freight	Pending	2026-06-15	NULL	5000.00	03-77881010

purchase_id	purchase_reference_num	invoice_id	invoice_number	supplier_id	supplier_reference_num
PO001	PO-EQP-001	INV011	INV-PO-001	S001	GLC-ENG-4821
PO002	PO-EQP-002	INV012	INV-PO-002	S002	GOV-LOG-1934
PO003	PO-EQP-003	INV013	INV-PO-003	S003	PVT-STE-7712
PO004	PO-EQP-004	INV014	INV-PO-004	S004	GLC-ENG-5603
PO005	PO-EQP-005	INV015	INV-PO-005	S005	CON-SHP-2291
PO006	PO-EQP-006	INV016	INV-PO-006	S006	GOV-MAR-8045
PO007	PO-EQP-007	INV017	INV-PO-007	S007	PVT-TEC-3906
PO008	PO-EQP-008	INV018	INV-PO-008	S008	GLC-SUB-1187
PO009	PO-EQP-009	INV019	INV-PO-009	S009	CON-OFF-6673
PO010	PO-EQP-010	INV020	INV-PO-010	S010	GOV-FLE-9052

19. company_certification

certification_id	certificate_serial_num	certificate_name	certificate_type	issue_date	expiry_date	renewal_date
CERT001	CERT-HSE-001	Occupational Safety Certification	Safety	2023-01-15	2026-01-15	2025-12-15
CERT002	CERT-ENG-002	Marine Engineering Certification	Technical	2022-06-10	2027-06-10	2027-05-10
CERT003	CERT-IT-003	Information Security Certification	Professional	2024-03-20	2027-03-20	2027-02-20
CERT004	CERT-FIN-004	Financial Compliance Certification	Compliance	2023-08-05	2026-08-05	2026-07-05
CERT005	CERT-OPS-005	Shipyard Operations Certification	Operational	2022-11-18	2027-11-18	2027-10-18
CERT006	CERT-PRO-006	Procurement Management Certification	Professional	2024-01-12	2027-01-12	2026-12-12
CERT007	CERT-ENG-007	Structural Inspection Certification	Technical	2023-04-08	2026-04-08	2026-03-08
CERT008	CERT-HR-008	Human Resource Management Certification	Professional	2024-02-01	2027-02-01	2026-12-01
CERT009	CERT-OPS-009	Maritime Safety Management Certification	Safety	2022-09-22	2027-09-22	2027-08-22
CERT010	CERT-FIN-010	Audit & Risk Certification	Compliance	2023-05-30	2026-05-30	2026-04-30

verification_date	verified_by	certificate_status	department_id	department_code
2025-12-20	Maritime Certification Board	Active	D001	ENG
2025-06-15	Maritime Certification Board	Active	D001	ENG
2025-03-25	IT Governance Council	Active	D003	IT
2025-07-10	Financial Standards Authority	Active	D004	FIN
2025-11-25	Marine Operations Council	Active	D005	OPS
2025-12-18	Procurement Standards Board	Active	D006	PROC
2025-03-15	Maritime Certification Board	Active	D001	ENG
2025-12-05	Human Resource Development Board	Active	D002	HR
2025-08-28	Marine Safety Authority	Active	D005	OPS
2025-05-05	Financial Standards Authority	Active	D004	FIN

20. payslip

payslip_id	payslip_code	scheduled_date	received_date	gross_salary	deductions	net_salary	salary_payment_date	account_number	payslip_status
PS001	PSL-ENG-001	2026-01-31	2026-01-30	8500.00	500.00	8000.00	2026-02-01	ACC-1001	Paid
PS002	PSL-FIN-002	2026-01-31	2026-01-30	4500.00	200.00	4300.00	2026-02-01	ACC-1002	Paid
PS003	PSL-IT-003	2026-01-31	2026-01-30	6200.00	300.00	5900.00	2026-02-01	ACC-1003	Paid
PS004	PSL-FIN-004	2026-01-31	2026-01-30	7800.00	400.00	7400.00	2026-02-01	ACC-1004	Paid
PS005	PSL-OPS-005	2026-01-31	2026-01-30	5000.00	250.00	4750.00	2026-02-01	ACC-1005	Paid
PS006	PSL-PROC-006	2026-01-31	2026-01-30	4800.00	200.00	4600.00	2026-02-01	ACC-1006	Paid
PS007	PSL-IT-007	2026-01-31	2026-01-30	6500.00	350.00	6150.00	2026-02-01	ACC-1007	Paid
PS008	PSL-HR-008	2026-01-31	2026-01-30	3200.00	100.00	3100.00	2026-02-01	ACC-1008	Paid
PS009	PSL-OPS-009	2026-01-31	2026-01-30	9200.00	600.00	8600.00	2026-02-01	ACC-1009	Paid
PS010	PSL-FIN-010	2026-01-31	2026-01-30	5500.00	300.00	5200.00	2026-02-01	ACC-1010	Paid

employee_id	employee_nric	department_id	department_code
2003150001	900101-14-5678	D001	ENG
1908120004	880212-10-3344	D004	FIN
2007060007	910305-08-7788	D003	IT
2139140011	850617-12-9900	D004	FIN
2304030005	920808-07-2211	D005	OPS
2111180006	870909-05-4455	D006	PROC
2201200003	930111-03-6677	D003	IT
2402010008	860223-09-8899	D002	HR
1806250009	900404-11-1122	D005	OPS
2209140010	940505-02-5566	D004	FIN

21. attendance

attendance_id	attendance_code	attendance_date	scheduled_checkin_time	scheduled_checkout_time	checkin_time	checkout_time	hours_worked	attendance_status	attendance_method
AT001	ATT-ENG-001	2026-06-01	08:00	17:00	07:55	17:10	9.25	Present	Biometric
AT002	ATT-FIN-002	2026-06-01	08:30	17:30	08:35	17:40	9.00	Present	Manual
AT003	ATT-IT-003	2026-06-01	08:00	17:00	08:10	17:05	8.75	Present	Biometric
AT004	ATT-OPS-004	2026-06-01	08:00	17:00	08:00	17:00	9.00	Present	RFID
AT005	ATT-PROC-005	2026-06-01	08:30	17:30	08:25	17:35	9.25	Present	Biometric
AT006	ATT-HR-006	2026-06-01	09:00	18:00	09:05	18:10	9.00	Present	Manual
AT007	ATT-IT-007	2026-06-01	08:00	17:00	08:15	17:20	8.75	Present	Biometric
AT008	ATT-HR-008	2026-06-01	09:00	18:00	09:00	18:00	9.00	Present	RFID
AT009	ATT-OPS-009	2026-06-01	08:00	17:00	08:05	17:00	8.75	Present	Biometric
AT010	ATT-FIN-010	2026-06-01	08:30	17:30	08:40	17:45	9.00	Present	Manual

employee_id	employee_nric	department_id	department_code
2003150001	900101-14-5678	D001	ENG
1908120004	880212-10-3344	D004	FIN
2007060007	910305-08-7788	D003	IT
2304030005	920808-07-2211	D005	OPS
2111180006	870909-05-4455	D006	PROC
2402010008	860223-09-8899	D002	HR
2201200003	930111-03-6677	D003	IT
2402010008	860223-09-8899	D002	HR
2304030005	900404-11-1122	D005	OPS
2402010008	940505-02-5566	D004	FIN

4.6 Data Layout – Explosion Table

1. contract_employee

contract_id	agreement_reference_number	employee_id	employee_nric
CT001	AGR-DEF-001	2003150001	900101-14-5678
CT002	AGR-MDE-002	1908120004	880212-10-3344
CT003	AGR-MRE-003	2007060007	910305-08-7788
CT004	AGR-DEF-004	2139140011	850617-12-9900
CT005	AGR-SHB-005	2304030005	920808-07-2211
CT006	AGR-CGD-006	2111180006	870909-05-4455
CT007	AGR-MTE-007	2201200003	930111-03-6677
CT008	AGR-MDE-008	2139140011	860223-09-8899
CT009	AGR-OFE-009	1806250009	900404-11-1122
CT010	AGR-MSE-010	2402010008	940505-02-5566

2. employee_purchase_order

employee_id	employee_nric	purchase_id	purchase_reference_num
2003150001	900101-14-5678	PO001	PO-EQP-001
1908120004	880212-10-3344	PO002	PO-EQP-002
2007060007	910305-08-7788	PO003	PO-EQP-003
2139140011	850617-12-9900	PO004	PO-EQP-004
2304030005	920808-07-2211	PO005	PO-EQP-005
2111180006	870909-05-4455	PO006	PO-EQP-006
2201200003	930111-03-6677	PO007	PO-EQP-007
2402010008	860223-09-8899	PO008	PO-EQP-008
1806250009	900404-11-1122	PO009	PO-EQP-009
2402010008	940505-02-5566	PO010	PO-EQP-010

3. employee_delivery

employee_id	employee_nric	delivery_id	tracking_number
2003150001	900101-14-5678	DEL001	TRK-LUN-001
1908120004	880212-10-3344	DEL002	TRK-LUN-002
2007060007	910305-08-7788	DEL003	TRK-LUN-003
2139140011	850617-12-9900	DEL004	TRK-LUN-004
2304030005	920808-07-2211	DEL005	TRK-LUN-005
2111180006	870909-05-4455	DEL006	TRK-LUN-006
2201200003	930111-03-6677	DEL007	TRK-LUN-007
2402010008	860223-09-8899	DEL008	TRK-LUN-008
1806250009	900404-11-1122	DEL009	TRK-LUN-009
2209140010	940505-02-5566	DEL010	TRK-LUN-010

4. work_order_employee

work_order_id	work_order_num	employee_id	employee_nric
WO001	WON-DEF-001	2003150001	900101-14-5678
WO002	WON-MDE-002	1908120004	880212-10-3344
WO003	WON-MRE-003	2007060007	910305-08-7788
WO004	WON-DEF-004	2139140011	850617-12-9900
WO005	WON-SHB-005	2304030005	920808-07-2211
WO006	WON-CGD-006	2111180006	870909-05-4455
WO007	WON-MTE-007	2201200003	930111-03-6677
WO008	WON-MDE-008	2402010008	860223-09-8899
WO009	WON-OFE-009	1806250009	900404-11-1122
WO010	WON-MSE-010	2209140010	940505-02-5566

5. invoice_employee

invoice_id	invoice_number	employee_id	employee_nric
INV001	INV-LUN-001	2003150001	900101-14-5678
INV002	INV-LUN-002	1908120004	880212-10-3344
INV003	INV-LUN-003	2007060007	910305-08-7788
INV004	INV-LUN-004	2139140011	850617-12-9900
INV005	INV-LUN-005	2304030005	920808-07-2211
INV006	INV-LUN-006	2111180006	870909-05-4455
INV007	INV-LUN-007	2201200003	930111-03-6677
INV008	INV-LUN-008	2402010008	860223-09-8899
INV009	INV-LUN-009	1806250009	900404-11-1122
INV010	INV-LUN-010	2209140010	940505-02-5566

6. vessel_employee

vessel_id	vessel_registration_num	employee_id	employee_nric
V001	VRN-LUN-001	2003150001	900101-14-5678
V002	VRN-LUN-002	2209140010	940505-02-5566
V003	VRN-LUN-003	2201200003	930111-03-6677
V004	VRN-LUN-004	1908120004	880212-10-3344
V005	VRN-LUN-005	2304030005	920808-07-2211
V006	VRN-LUN-006	2111180006	870909-05-4455
V007	VRN-LUN-007	2007060007	910305-08-7788
V008	VRN-LUN-008	2402010008	850617-12-9900
V009	VRN-LUN-009	1806250009	900404-11-1122
V010	VRN-LUN-010	2402010008	860223-09-8899

7. employee_company_certification

employee_id	employee_nric	certification_id	certificate_serial_num
2003150001	900101-14-5678	CERT001	CERT-HSE-001
1908120004	880212-10-3344	CERT002	CERT-ENG-002
2007060007	910305-08-7788	CERT003	CERT-IT-003
2139140011	850617-12-9900	CERT004	CERT-FIN-004
2304030005	920808-07-2211	CERT005	CERT-OPS-005
2111180006	870909-05-4455	CERT006	CERT-PRO-006
2201200003	930111-03-6677	CERT007	CERT-ENG-007
2402010008	860223-09-8899	CERT008	CERT-HR-008
1806250009	900404-11-1122	CERT009	CERT-OPS-009
2209140010	940505-02-5566	CERT010	CERT-FIN-010

8. employee_service_request

employee_id	employee_nric	request_id	request_code
2003150001	900101-14-5678	R001	REQ-DEF-001
1908120004	880212-10-3344	R002	REQ-MDE-002
2007060007	910305-08-7788	R003	REQ-MRE-003
2139140011	850617-12-9900	R004	REQ-DEF-004
2304030005	920808-07-2211	R005	REQ-SHB-005
2111180006	870909-05-4455	R006	REQ-CGD-006
2201200003	930111-03-6677	R007	REQ-MTE-007
2402010008	860223-09-8899	R008	REQ-MDE-008
1806250009	900404-11-1122	R009	REQ-OFE-009
2209140010	940505-02-5566	R010	REQ-MSE-010

9. department_work_order

department_id	department_code	work_order_id	work_order_num
D001	ENG	WO001	WON-DEF-001
D002	HR	WO002	WON-MDE-002
D003	IT	WO003	WON-MRE-003
D004	FIN	WO004	WON-DEF-004
D005	OPS	WO005	WON-SHB-005
D006	PROC	WO006	WON-CGD-006
D007	QAQC	WO007	WON-MTE-007
D008	LOG	WO008	WON-MDE-008
D009	RND	WO009	WON-OFE-009
D010	LEG	WO010	WON-MSE-010

10. external_party_department

party_id	registration_number	department_id	department_code
P001	REG-NUM-1901001	D001	ENG
P002	REG-NUM-2012002	D002	HR
P003	REG-NUM-1812003	D003	IT
P004	REG-NUM-2112004	D004	FIN
P005	REG-NUM-1711005	D005	OPS
P006	REG-NUM-2202006	D006	PROC
P007	REG-NUM-1908007	D007	QAQC
P008	REG-NUM-2006008	D008	LOG
P009	REG-NUM-1804009	D009	RND
P010	REG-NUM-2307010	D010	LEG

11. external_party_employee

party_id	registration_number	employee_id	employee_nric
P001	REG-NUM-1901001	2003150001	900101-14-5678
P002	REG-NUM-2012002	1908120004	880212-10-3344
P003	REG-NUM-1812003	2007060007	910305-08-7788
P004	REG-NUM-2112004	2139140011	850617-12-9900
P005	REG-NUM-1711005	2304030005	920808-07-2211
P006	REG-NUM-2202006	2111180006	870909-05-4455
P007	REG-NUM-1908007	2201200003	930111-03-6677
P008	REG-NUM-2006008	2402010008	860223-09-8899
P009	REG-NUM-1804009	1806250009	900404-11-1122
P010	REG-NUM-2307010	2209140010	940505-02-5566

12. material_delivery

material_item_id	material_code	delivery_id	tracking_number
MAT001	STL-A36	DEL001	TRK-LUN-001
MAT002	ENG-GR	DEL002	TRK-LUN-002
MAT003	RAD-MOD	DEL003	TRK-LUN-003
MAT004	HYD-PMP	DEL004	TRK-LUN-004
MAT005	CAB-MAR	DEL005	TRK-LUN-005
MAT006	CLG-UNIT	DEL006	TRK-LUN-006
MAT007	PRP-BLD	DEL007	TRK-LUN-007
MAT008	NAV-SEN	DEL008	TRK-LUN-008
MAT009	FUL-PMP	DEL009	TRK-LUN-009
MAT010	SAF-HLM	DEL010	TRK-LUN-010

13. employee_project

employee_id	employee_nric	project_id	project_reference_num
2003150001	900101-14-5678	PR001	PROJ-DEF-001
1908120004	880212-10-3344	PR002	PROJ-MDE-002
2007060007	910305-08-7788	PR003	PROJ-MRE-003
2139140011	850617-12-9900	PR004	PROJ-DEF-004
2304030005	920808-07-2211	PR005	PROJ-SHB-005
2111180006	870909-05-4455	PR006	PROJ-CGD-006
2201200003	930111-03-6677	PR007	PROJ-MTE-007
2402010008	860223-09-8899	PR008	PROJ-MDE-008
1806250009	900404-11-1122	PR009	PROJ-OFE-009
2209140010	940505-02-5566	PR010	PROJ-MSE-010

14. purchase_order_material

purchase_id	purchase_reference_num	material_item_id	material_code
PO001	PO-EQP-001	MAT001	STL-A36
PO002	PO-EQP-002	MAT002	ENG-GR
PO003	PO-EQP-003	MAT003	RAD-MOD
PO004	PO-EQP-004	MAT004	HYD-PMP
PO005	PO-EQP-005	MAT005	CAB-MAR
PO006	PO-EQP-006	MAT006	CLG-UNIT
PO007	PO-EQP-007	MAT007	PRP-BLD
PO008	PO-EQP-008	MAT008	NAV-SEN
PO009	PO-EQP-009	MAT009	FUL-PMP
PO010	PO-EQP-010	MAT010	SAF-HLM

15. project_vessel

project_id	project_reference_num	vessel_id	vessel_registration_num
PR001	PROJ-DEF-001	V001	VRN-LUN-001
PR002	PROJ-MDE-002	V002	VRN-LUN-002
PR003	PROJ-MRE-003	V003	VRN-LUN-003
PR004	PROJ-DEF-004	V004	VRN-LUN-004
PR005	PROJ-SHB-005	V005	VRN-LUN-005
PR006	PROJ-CGD-006	V006	VRN-LUN-006
PR007	PROJ-MTE-007	V007	VRN-LUN-007
PR008	PROJ-MDE-008	V008	VRN-LUN-008
PR009	PROJ-OFE-009	V009	VRN-LUN-009
PR010	PROJ-MSE-010	V010	VRN-LUN-010

16. department_service_request

department_id	department_code	request_id	request_code
D001	ENG	R001	REQ-DEF-001
D002	HR	R002	REQ-MDE-002
D003	IT	R003	REQ-MRE-003
D004	FIN	R004	REQ-DEF-004
D005	OPS	R005	REQ-SHB-005
D006	PROC	R006	REQ-CGD-006
D007	QAQC	R007	REQ-MTE-007
D008	LOG	R008	REQ-MDE-008
D009	RND	R009	REQ-OFE-009
D010	LEG	R010	REQ-MSE-010

17. project_company_certification

project_id	project_reference_num	certification_id	certification_serial_num
PR001	PROJ-DEF-001	CERT001	CERT-HSE-001
PR002	PROJ-MDE-002	CERT002	CERT-ENG-002
PR003	PROJ-MRE-003	CERT003	CERT-IT-003
PR004	PROJ-DEF-004	CERT004	CERT-FIN-004
PR005	PROJ-SHB-005	CERT005	CERT-OPS-005
PR006	PROJ-CGD-006	CERT006	CERT-PRO-006
PR007	PROJ-MTE-007	CERT007	CERT-ENG-007
PR008	PROJ-MDE-008	CERT008	CERT-HR-008
PR009	PROJ-OFE-009	CERT009	CERT-OPS-009
PR010	PROJ-MSE-010	CERT010	CERT-FIN-010

18. invoice_payment

invoice_id	invoice_number	payment_id	payment_reference_num
INV001	INV-LUN-001	PAY001	PAY-COM-001
INV002	INV-LUN-002	PAY002	PAY-PAR-002
INV003	INV-LUN-003	PAY003	PAY-COM-003
INV004	INV-LUN-004	PAY004	PAY-PAR-004
INV005	INV-LUN-005	PAY005	PAY-COM-005
INV006	INV-LUN-006	PAY006	PAY-COM-006
INV007	INV-LUN-007	PAY007	PAY-COM-007
INV008	INV-LUN-008	PAY008	PAY-PAR-008
INV009	INV-LUN-009	PAY009	PAY-COM-009
INV010	INV-LUN-010	PAY010	PAY-PAR-010

19. employee_payment

employee_id	employee_nric	payment_id	payment_reference_num
2003150001	900101-14-5678	PAY001	PAY-COM-001
1908120004	880212-10-3344	PAY002	PAY-PAR-002
2007060007	910305-08-7788	PAY003	PAY-COM-003
2139140011	850617-12-9900	PAY004	PAY-PAR-004
2304030005	920808-07-2211	PAY005	PAY-COM-005
2111180006	870909-05-4455	PAY006	PAY-COM-006
2201200003	930111-03-6677	PAY007	PAY-COM-007
2402010008	860223-09-8899	PAY008	PAY-PAR-008
1806250009	900404-11-1122	PAY009	PAY-COM-009
2209140010	940505-02-5566	PAY010	PAY-PAR-010

20. payment_project

payment_id	payment_reference_num	project_id	project_reference_num
PAY001	PAY-COM-001	PR001	PROJ-DEF-001
PAY002	PAY-PAR-002	PR002	PROJ-MDE-002
PAY003	PAY-COM-003	PR003	PROJ-MRE-003
PAY004	PAY-PAR-004	PR004	PROJ-DEF-004
PAY005	PAY-COM-005	PR005	PROJ-SHB-005
PAY006	PAY-COM-006	PR006	PROJ-CGD-006
PAY007	PAY-COM-007	PR007	PROJ-MTE-007
PAY008	PAY-PAR-008	PR008	PROJ-MDE-008
PAY009	PAY-COM-009	PR009	PROJ-OFE-009
PAY010	PAY-PAR-010	PR010	PROJ-MSE-010

21. equipment_project

equipment_id	equipment_serial_num	project_id	project_reference_num
EQ001	EQ-LUN-001-STEEL	PR001	PROJ-DEF-001
EQ002	EQ-LUN-002-ENG	PR002	PROJ-MDE-002
EQ003	EQ-LUN-003-RAD	PR003	PROJ-MRE-003
EQ004	EQ-LUN-004-HYD	PR004	PROJ-DEF-004
EQ005	EQ-LUN-005-CAB	PR005	PROJ-SHB-005
EQ006	EQ-LUN-006-CLG	PR006	PROJ-CGD-006
EQ007	EQ-LUN-007-PRP	PR007	PROJ-MTE-007
EQ008	EQ-LUN-008-NAV	PR008	PROJ-MDE-008
EQ009	EQ-LUN-009-FUL	PR009	PROJ-OFE-009
EQ010	EQ-LUN-010-SAF	PR010	PROJ-MSE-010

22. department_vessel

department_id	department_code	vessel_id	vessel_registration_number
D001	ENG	V001	VRN-LUN-001
D002	HR	V002	VRN-LUN-002
D003	IT	V003	VRN-LUN-003
D004	FIN	V004	VRN-LUN-004
D005	OPS	V005	VRN-LUN-005
D006	PROC	V006	VRN-LUN-006
D007	QAQC	V007	VRN-LUN-007
D008	LOG	V008	VRN-LUN-008
D009	RND	V009	VRN-LUN-009
D010	LEG	V010	VRN-LUN-010

23. department_work_report

department_id	department_code	report_id	report_reference_num
D001	ENG	RPT001	REP-MNT-001
D002	HR	RPT002	REP-INS-002
D003	IT	RPT003	REP-PRG-003
D004	FIN	RPT004	REP-HSE-004
D005	OPS	RPT005	REP-TEC-005
D006	PROC	RPT006	REP-MNT-006
D007	QAQC	RPT007	REP-OPR-007
D008	LOG	RPT008	REP-ENG-008
D009	RND	RPT009	REP-LOG-009
D010	LEG	RPT010	REP-STR-010

24. purchase_order_payment

purchase_id	purchase_reference_num	payment_id	payment_reference_num
PO001	PO-EQP-001	PAY001	PAY-COM-001
PO002	PO-EQP-002	PAY002	PAY-PAR-002
PO003	PO-EQP-003	PAY003	PAY-COM-003
PO004	PO-EQP-004	PAY004	PAY-PAR-004
PO005	PO-EQP-005	PAY005	PAY-COM-005
PO006	PO-EQP-006	PAY006	PAY-COM-006
PO007	PO-EQP-007	PAY007	PAY-COM-007
PO008	PO-EQP-008	PAY008	PAY-PAR-008
PO009	PO-EQP-009	PAY009	PAY-COM-009
PO010	PO-EQP-010	PAY010	PAY-PAR-010

25. supplier_material

supplier_id	supplier_reference_num	material_item_id	material_code
S001	GLC-ENG-4821	MAT001	STL-A36
S002	GOV-LOG-1934	MAT002	ENG-GR
S003	PVT-STE-7712	MAT003	RAD-MOD
S004	GLC-ENG-5603	MAT004	HYD-PMP
S005	CON-SHP-2291	MAT005	CAB-MAR
S006	GOV-MAR-8045	MAT006	CLG-UNIT
S007	PVT-TEC-3906	MAT007	PRP-BLD
S008	GLC-SUB-1187	MAT008	NAV-SEN
S009	CON-OFF-6673	MAT009	FUL-PMP
S010	GOV-FLE-9052	MAT010	SAF-HLM

26. purchase_order_department

purchase_id	purchase_reference_num	department_id	department_code
PO001	PO-EQP-001	D001	ENG
PO002	PO-EQP-002	D002	HR
PO003	PO-EQP-003	D003	IT
PO004	PO-EQP-004	D004	FIN
PO005	PO-EQP-005	D005	OPS
PO006	PO-EQP-006	D006	PROC
PO007	PO-EQP-007	D007	QAQC
PO008	PO-EQP-008	D008	LOG
PO009	PO-EQP-009	D009	RND
PO010	PO-EQP-010	D010	LEG

27. invoice-department

invoice_id	invoice_number	department_id	department_code
INV001	INV-LUN-001	D001	ENG
INV002	INV-LUN-002	D002	HR
INV003	INV-LUN-003	D003	IT
INV004	INV-LUN-004	D004	FIN
INV005	INV-LUN-005	D005	OPS
INV006	INV-LUN-006	D006	PROC
INV007	INV-LUN-007	D007	QAQC
INV008	INV-LUN-008	D008	LOG
INV009	INV-LUN-009	D009	RND
INV010	INV-LUN-010	D010	LEG

28. material_project

material_item_id	material_code	project_id	project_reference_num
MAT001	STL-A36	PR001	PROJ-DEF-001
MAT002	ENG-GR	PR002	PROJ-MDE-002
MAT003	RAD-MOD	PR003	PROJ-MRE-003
MAT004	HYD-PMP	PR004	PROJ-DEF-004
MAT005	CAB-MAR	PR005	PROJ-SHB-005
MAT006	CLG-UNIT	PR006	PROJ-CGD-006
MAT007	PRP-BLD	PR007	PROJ-MTE-007
MAT008	NAV-SEN	PR008	PROJ-MDE-008
MAT009	FUL-PMP	PR009	PROJ-OFE-009
MAT010	SAF-HLM	PR010	PROJ-MSE-010

29. employee_material

employee_id	employee_nric	material_item_id	material_code
2003150001	900101-14-5678	MAT001	STL-A36
2209140010	940505-02-5566	MAT002	ENG-GR
2201200003	930111-03-6677	MAT003	RAD-MOD
1908120004	880212-10-3344	MAT004	HYD-PMP
2304030005	920808-07-2211	MAT005	CAB-MAR
2111180006	870909-05-4455	MAT006	CLG-UNIT
2007060007	910305-08-7788	MAT007	PRP-BLD
2139140011	850617-12-9900	MAT008	NAV-SEN
1806250009	900404-11-1122	MAT009	FUL-PMP
2402010008	860223-09-8899	MAT010	SAF-HLM

30. employee_equipment

employee_id	employee_nric	equipment_id	equipment_serial_num
2003150001	900101-14-5678	EQ001	EQ-LUN-001-STEEL
2209140010	940505-02-5566	EQ002	EQ-LUN-002-ENG
2201200003	930111-03-6677	EQ003	EQ-LUN-003-RAD
1908120004	880212-10-3344	EQ004	EQ-LUN-004-HYD
2304030005	920808-07-2211	EQ005	EQ-LUN-005-CAB
2111180006	870909-05-4455	EQ006	EQ-LUN-006-CLG
2007060007	910305-08-7788	EQ007	EQ-LUN-007-PRP
2139140011	850617-12-9900	EQ008	EQ-LUN-008-NAV
1806250009	900404-11-1122	EQ009	EQ-LUN-009-FUL
2402010008	860223-09-8899	EQ010	EQ-LUN-010-SAF

31. equipment_vessel

equipment_id	equipment_serial_num	vessel_id	vessel_registration_number
EQ001	EQ-LUN-001-STEEL	V001	VRN-LUN-001
EQ002	EQ-LUN-002-ENG	V002	VRN-LUN-002
EQ003	EQ-LUN-003-RAD	V003	VRN-LUN-003
EQ004	EQ-LUN-004-HYD	V004	VRN-LUN-004
EQ005	EQ-LUN-005-CAB	V005	VRN-LUN-005
EQ006	EQ-LUN-006-CLG	V006	VRN-LUN-006
EQ007	EQ-LUN-007-PRP	V007	VRN-LUN-007
EQ008	EQ-LUN-008-NAV	V008	VRN-LUN-008
EQ009	EQ-LUN-009-FUL	V009	VRN-LUN-009
EQ010	EQ-LUN-010-SAF	V010	VRN-LUN-010

32. material_vessel

material_item_id	material_code	vessel_id	vessel_registration_number
MAT001	STL-A36	V001	VRN-LUN-001
MAT002	ENG-GR	V002	VRN-LUN-002
MAT003	RAD-MOD	V003	VRN-LUN-003
MAT004	HYD-PMP	V004	VRN-LUN-004
MAT005	CAB-MAR	V005	VRN-LUN-005
MAT006	CLG-UNIT	V006	VRN-LUN-006
MAT007	PRP-BLD	V007	VRN-LUN-007
MAT008	NAV-SEN	V008	VRN-LUN-008
MAT009	FUL-PMP	V009	VRN-LUN-009
MAT010	SAF-HLM	V010	VRN-LUN-010

33. equipment_work_order

equipment_id	equipment_serial_num	work_order_id	work_order_num
EQ001	EQ-LUN-001-STEEL	WO001	WON-DEF-001
EQ002	EQ-LUN-002-ENG	WO002	WON-MDE-002
EQ003	EQ-LUN-003-RAD	WO003	WON-MRE-003
EQ004	EQ-LUN-004-HYD	WO004	WON-DEF-004
EQ005	EQ-LUN-005-CAB	WO005	WON-SHB-005
EQ006	EQ-LUN-006-CLG	WO006	WON-CGD-006
EQ007	EQ-LUN-007-PRP	WO007	WON-MTE-007
EQ008	EQ-LUN-008-NAV	WO008	WON-MDE-008
EQ009	EQ-LUN-009-FUL	WO009	WON-OFE-009
EQ010	EQ-LUN-010-SAF	WO010	WON-MSE-010

34. material_work_order

material_item_id	material_code	work_order_id	work_order_num
MAT001	STL-A36	WO001	WON-DEF-001
MAT002	ENG-GR	WO002	WON-MDE-002
MAT003	RAD-MOD	WO003	WON-MRE-003
MAT004	HYD-PMP	WO004	WON-DEF-004
MAT005	CAB-MAR	WO005	WON-SHB-005
MAT006	CLG-UNIT	WO006	WON-CGD-006
MAT007	PRP-BLD	WO007	WON-MTE-007
MAT008	NAV-SEN	WO008	WON-MDE-008
MAT009	FUL-PMP	WO009	WON-OFE-009
MAT010	SAF-HLM	WO010	WON-MSE-010

35. equipment_work_report

equipment_id	equipment_serial_num	report_id	report_reference_num
EQ001	EQ-LUN-001-STEEL	RPT001	REP-MNT-001
EQ002	EQ-LUN-002-ENG	RPT002	REP-INS-002
EQ003	EQ-LUN-003-RAD	RPT003	REP-PRG-003
EQ004	EQ-LUN-004-HYD	RPT004	REP-HSE-004
EQ005	EQ-LUN-005-CAB	RPT005	REP-TEC-005
EQ006	EQ-LUN-006-CLG	RPT006	REP-MNT-006
EQ007	EQ-LUN-007-PRP	RPT007	REP-OPR-007
EQ008	EQ-LUN-008-NAV	RPT008	REP-ENG-008
EQ009	EQ-LUN-009-FUL	RPT009	REP-LOG-009
EQ010	EQ-LUN-010-SAF	RPT010	REP-STR-010

36. material_work_report

material_item_id	material_code	report_id	report_reference_num
MAT001	STL-A36	RPT001	REP-MNT-001
MAT002	ENG-GR	RPT002	REP-INS-002
MAT003	RAD-MOD	RPT003	REP-PRG-003
MAT004	HYD-PMP	RPT004	REP-HSE-004
MAT005	CAB-MAR	RPT005	REP-TEC-005
MAT006	CLG-UNIT	RPT006	REP-MNT-006
MAT007	PRP-BLD	RPT007	REP-OPR-007
MAT008	NAV-SEN	RPT008	REP-ENG-008
MAT009	FUL-PMP	RPT009	REP-LOG-009
MAT010	SAF-HLM	RPT010	REP-STR-010

37. material_contract

material_item_id	material_code	contract_id	agreement_reference_number
MAT001	STL-A36	CT001	AGR-DEF-001
MAT002	ENG-GR	CT002	AGR-MDE-002
MAT003	RAD-MOD	CT003	AGR-MRE-003
MAT004	HYD-PMP	CT004	AGR-DEF-004
MAT005	CAB-MAR	CT005	AGR-SHB-005
MAT006	CLG-UNIT	CT006	AGR-CGD-006
MAT007	PRP-BLD	CT007	AGR-MTE-007
MAT008	NAV-SEN	CT008	AGR-MDE-008
MAT009	FUL-PMP	CT009	AGR-OFE-009
MAT010	SAF-HLM	CT010	AGR-MSE-010

38. equipment_contract

equipment_id	equipment_serial_num	contract_id	agreement_reference_number
EQ001	EQ-LUN-001-STEEL	CT001	AGR-DEF-001
EQ002	EQ-LUN-002-ENG	CT002	AGR-MDE-002
EQ003	EQ-LUN-003-RAD	CT003	AGR-MRE-003
EQ004	EQ-LUN-004-HYD	CT004	AGR-DEF-004
EQ005	EQ-LUN-005-CAB	CT005	AGR-SHB-005
EQ006	EQ-LUN-006-CLG	CT006	AGR-CGD-006
EQ007	EQ-LUN-007-PRP	CT007	AGR-MTE-007
EQ008	EQ-LUN-008-NAV	CT008	AGR-MDE-008
EQ009	EQ-LUN-009-FUL	CT009	AGR-OFE-009
EQ010	EQ-LUN-010-SAF	CT010	AGR-MSE-010

39. invoice_material

invoice_id	invoice_number	material_item_id	material_code
INV001	INV-LUN-001	MAT001	STL-A36
INV002	INV-LUN-002	MAT002	ENG-GR
INV003	INV-LUN-003	MAT003	RAD-MOD
INV004	INV-LUN-004	MAT004	HYD-PMP
INV005	INV-LUN-005	MAT005	CAB-MAR
INV006	INV-LUN-006	MAT006	CLG-UNIT
INV007	INV-LUN-007	MAT007	PRP-BLD
INV008	INV-LUN-008	MAT008	NAV-SEN
INV009	INV-LUN-009	MAT009	FUL-PMP
INV010	INV-LUN-010	MAT010	SAF-HLM

40. payment_equipment

payment_id	payment_reference_num	equipment_id	equipment_serial_num
PAY001	PAY-COM-001	EQ001	EQ-LUN-001-STEEL
PAY002	PAY-PAR-002	EQ002	EQ-LUN-002-ENG
PAY003	PAY-COM-003	EQ003	EQ-LUN-003-RAD
PAY004	PAY-COM-004	EQ004	EQ-LUN-004-HYD
PAY005	PAY-PAR-005	EQ005	EQ-LUN-005-CAB
PAY006	PAY-COM-006	EQ006	EQ-LUN-006-CLG
PAY007	PAY-PAR-007	EQ007	EQ-LUN-007-PRP
PAY008	PAY-PAR-008	EQ008	EQ-LUN-008-NAV
PAY009	PAY-COM-009	EQ009	EQ-LUN-009-FUL
PAY010	PAY-PAR-010	EQ010	EQ-LUN-010-SAF

41. delivery_payment

delivery_id	tracking_number	payment_id	payment_reference_num
DEL001	TRK-LUN-001	PAY001	PAY-COM-001
DEL002	TRK-LUN-002	PAY002	PAY-PAR-002
DEL003	TRK-LUN-003	PAY003	PAY-COM-003
DEL004	TRK-LUN-004	PAY004	PAY-COM-004
DEL005	TRK-LUN-005	PAY005	PAY-PAR-005
DEL006	TRK-LUN-006	PAY006	PAY-COM-006
DEL007	TRK-LUN-007	PAY007	PAY-PAR-007
DEL008	TRK-LUN-008	PAY008	PAY-PAR-008
DEL009	TRK-LUN-009	PAY009	PAY-COM-009
DEL010	TRK-LUN-010	PAY010	PAY-PAR-010

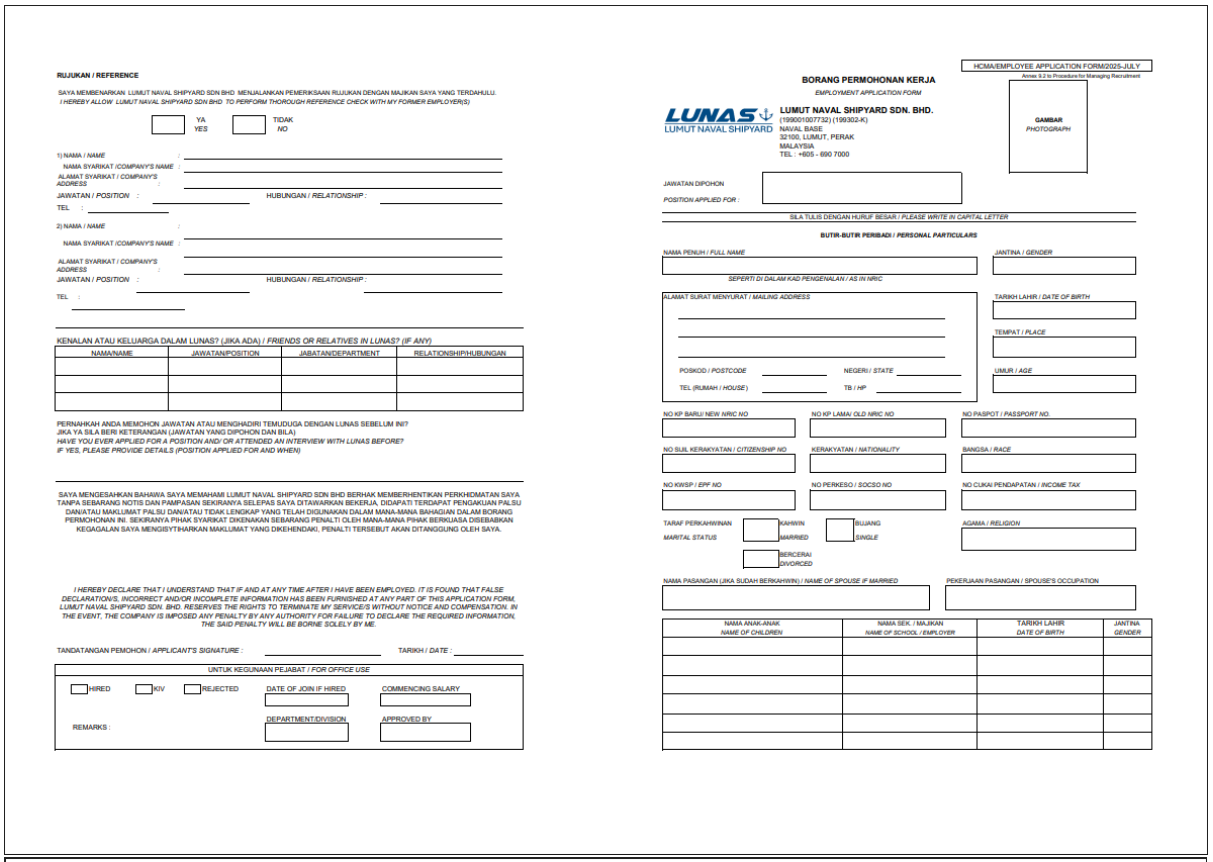
42. employee_supervisor

supervisor_employee_id	supervisor_employee_nric	supervisor_employee_id	supervisor_employee_nric
1806250009	900404-11-1122	2003150001	900101-14-5678
1806250009	900404-11-1122	2209140010	940505-02-5566
1806250009	900404-11-1122	2201200003	930111-03-6677
1806250009	900404-11-1122	1908120004	880212-10-3344
1806250009	900404-11-1122	2304030005	920808-07-2211
1806250009	900404-11-1122	2111180006	870909-05-4455
1806250009	900404-11-1122	2007060007	910305-08-7788
1806250009	900404-11-1122	2139140011	850617-12-9900
1806250009	900404-11-1122	1806250009	900404-11-1122
1806250009	900404-11-1122	2402010008	860223-09-8899

43. payment_material

payment_id	payment_reference_num	material_item_id	material_code
PAY001	PAY-COM-001	MAT001	STL-A36
PAY002	PAY-PAR-002	MAT002	ENG-GR
PAY003	PAY-COM-003	MAT003	RAD-MOD
PAY004	PAY-PAR-004	MAT004	HYD-PMP
PAY005	PAY-COM-005	MAT005	CAB-MAR
PAY006	PAY-COM-006	MAT006	CLG-UNIT
PAY007	PAY-COM-007	MAT007	PRP-BLD
PAY008	PAY-PAR-008	MAT008	NAV-SEN
PAY009	PAY-COM-009	MAT009	FUL-PMP
PAY010	PAY-PAR-010	MAT010	SAF-HLM

4.7 Blank Forms



RUJUKAN / REFERENCE

SAYA MEMBENARKAN LUMUT NAVAL SHIPYARD SDN BHD MENJALANKAN PEMERIKSAAN RUJUKAN DENGAN MAJIKAN SAYA YANG TERDAHULU.
I HEREBY ALLOW LUMUT NAVAL SHIPYARD SDN BHD TO PERFORM THOROUGH REFERENCE CHECK WITH MY FORMER EMPLOYER(S)

☐ YA *YES* ☐ TIDAK *NO*

1) NAMA / *NAME* :
NAMA SYARIKAT /*COMPANY'S NAME* :
ALAMAT SYARIKAT / *COMPANY'S ADDRESS*
JAWATAN / *POSITION* : HUBUNGAN / *RELATIONSHIP* :
TEL :

2) NAMA / *NAME* :
NAMA SYARIKAT /*COMPANY'S NAME* :
ALAMAT SYARIKAT / *COMPANY'S ADDRESS*
JAWATAN / *POSITION* : HUBUNGAN / *RELATIONSHIP* :
TEL :

KENALAN ATAU KELUARGA DALAM LUNAS? (JIKA ADA) / *FRIENDS OR RELATIVES IN LUNAS? (IF ANY)*

NAMA/NAME	JAWATAN/POSITION	JABATAN/DEPARTMENT	RELATION/SHIP/HUBUNGAN

PERNAHKAH ANDA MEMOHON JAWATAN ATAU MENGHADIRI TEMUDUGA DENGAN LUNAS SEBELUM INI?
JIKA YA SILA BERI KETERANGAN (JAWATAN YANG DIPOHON DAN BILA)
HAVE YOU EVER APPLIED FOR A POSITION AND/ OR ATTENDED AN INTERVIEW WITH LUNAS BEFORE?
IF YES, PLEASE PROVIDE DETAILS (POSITION APPLIED FOR AND WHEN)

SAYA MENGESAHKAN BAHAWA SAYA MEMAHAMI LUMUT NAVAL SHIPYARD SDN BHD BERHAK MEMBERHENTIKAN PERKHIDMATAN SAYA TANPA SEBARANG NOTIS DAN PAMPASAN SEKIRANYA SELEPAS SAYA DITAWARKAN BEKERJA, DIDAPATI TERDAPAT PENGAKUAN PALSU DAN/ATAU MAKLUMAT PALSU DAN/ATAU TIDAK LENGKAP YANG TELAH DIGUNAKAN DALAM MANA-MANA BAHAGIAN DALAM BORANG PERMOHONAN INI. SEKIRANYA PIHAK SYARIKAT DIKENAKAN SEBARANG PENALTI OLEH MANA-MANA PIHAK BERKUASA DISEBABKAN KEGAGALAN SAYA MENGISYTIHARKAN MAKLUMAT YANG DIKEHENDAKI, PENALTI TERSEBUT AKAN DITANGGUNG OLEH SAYA.

I HEREBY DECLARE THAT I UNDERSTAND THAT IF AND AT ANY TIME AFTER I HAVE BEEN EMPLOYED, IT IS FOUND THAT FALSE DECLARATION/S, INCORRECT AND/OR INCOMPLETE INFORMATION HAS BEEN FURNISHED AT ANY PART OF THIS APPLICATION FORM, LUMUT NAVAL SHIPYARD SDN. BHD. RESERVES THE RIGHTS TO TERMINATE MY SERVICE/S WITHOUT NOTICE AND COMPENSATION. IN THE EVENT, THE COMPANY IS IMPOSED ANY PENALTY BY ANY AUTHORITY FOR FAILURE TO DECLARE THE REQUIRED INFORMATION, THE SAID PENALTY WILL BE BORNE SOLELY BY ME.

TANDATANGAN PEMOHON / *APPLICANT'S SIGNATURE* : TARIKH / *DATE* :

UNTUK KEGUNAAN PEJABAT / *FOR OFFICE USE*

☐ HIRED ☐ KIV ☐ REJECTED DATE OF JOIN IF HIRED COMMENCING SALARY
REMARKS : DEPARTMENT/DIVISION APPROVED BY

HCMA/EMPLOYEE APPLICATION FORM/2025-JULY
Annex 9.2 to Procedure for Managing Recruitment

BORANG PERMOHONAN KERJA
EMPLOYMENT APPLICATION FORM

LUMUT NAVAL SHIPYARD SDN. BHD.
(199001007732) (199302-K)
NAVAL BASE
32100, LUMUT, PERAK
MALAYSIA
TEL : +605 - 690 7000

GAMBAR
PHOTOGRAPH

JAWATAN DIPOHON
POSITION APPLIED FOR :

SILA TULIS DENGAN HURUF BESAR / *PLEASE WRITE IN CAPITAL LETTER*

BUTIR-BUTIR PERIBADI / *PERSONAL PARTICULARS*

NAMA PENUH / *FULL NAME* JANTINA / *GENDER*
SEPERTI DI DALAM KAD PENGENALAN / AS IN NRIC

ALAMAT SURAT MENYURAT / *MAILING ADDRESS* TARIKH LAHIR / *DATE OF BIRTH*
TEMPAT / *PLACE*
POSKOD / *POSTCODE* NEGERI/ *STATE* UMUR / *AGE*
TEL (RUMAH / *HOUSE*) TB / *HP*

NO KP BARU/ *NEW NRIC NO* NO KP LAMA/ *OLD NRIC NO* NO PASPOT / *PASSPORT NO.*
NO SIJIL KERAKYATAN / *CITIZENSHIP NO* KERAKYATAN / *NATIONALITY* BANGSA / *RACE*
NO KWSP / *EPF NO* NO PERKESO / *SOCSO NO* NO CUKAI/ PENDAPATAN / *INCOME TAX*

TARAF PERKAHWINAN / *MARITAL STATUS* ☐ KAHWIN *MARRIED* ☐ BUJANG *SINGLE* ☐ BERCERAI *DIVORCED*
AGAMA / *RELIGION*

NAMA PASANGAN (JIKA SUDAH BERKAHWIN) / *NAME OF SPOUSE IF MARRIED* PEKERJAAN PASANGAN / *SPOUSE'S OCCUPATION*

NAMA ANAK-ANAK / *NAME OF CHILDREN*	NAMA SEK. / MAJIKAN / *NAME OF SCHOOL / EMPLOYER*	TARIKH LAHIR / *DATE OF BIRTH*	JANTINA / *GENDER*

NAMA ADIK BERADIK / *NAME OF SIBLINGS*	NAMA SEK. / MAJIKAN / *NAME OF SCHOOL / EMPLOYER*	TARIKH LAHIR / *DATE OF BIRTH*	JANTINA / *GENDER*

PENDIDIKAN (KEPILKAN SALINAN SIJIL)
EDUCATION (ATTACH COPIES OF CERTIFICATE)

NAMA SEKOLAH / INSTITUSI / *NAME OF SCHOOL/INSTITUTION*	TAHUN (YEAR) MULA (START)	TAHUN (YEAR) AKHIR (END)	KELAYAKAN / MAJOR / *QUALIFICATION / MAJOR*	KEPUTUSAN / *RESULT*

KURSUS-KURSUS ATAU LATIHAN KHAS YANG SEDANG ATAU TELAH DIKUTI
SPECIAL COURSES OR TRAINING COMPLETED OR CURRENTLY PURSUED

NAMA KURSUS / LATIHAN / *NAME OF TRAINING*	NAMA KOLEJ /UNIVERSITI / NAME OF COLLEGE/UNIVERSITY	TARIKH (DATE) MULA (START)	TARIKH (DATE) AKHIR (END)	TARAF KELULUSAN / QUALIFICATION LEVEL

PENGALAMAN KERJA
WORKING EXPERIENCE

NAMA MAJIKAN / *EMPLOYER NAME*	DARI / *FROM*	HINGGA / *TO*	GAJI KASAR (MULA) / *BASIC SALARY*	GAJI KASAR (AKHIR) / BASIC SALARY	JAWATAN AKHIR / *LAST POSITION*	SEBAB BERHENTI / REASON FOR LEAVING

KEMAHIRAN BAHASA & LOGHAT
PROFICIENCY IN LANGUAGES & DIALECTS

BAHASA & LOGHAT / *LANGUAGE & DIALECTS*	MEMADAI / *ADEQUATE*	SEDERHANA / *AVERAGE*	BAIK / *GOOD*

BON - ADAKAH ANDA TERIKAT DENGAN PERJANJIAN PERKHIDMATAN DENGAN MANA - MANA JABATAN ATAU PERTUBUHAN?
BOND - ARE YOU BOUND TO SERVE ANY DEPARTMENT OR ORGANISATION?
TERANGKAN SECARA RINGKAS / *DESCRIBE BRIEFLY:*

MINAT,AKTIVITI, SUKAN, HOBI, KEAHLIAN KELAB DAN LAIN-LAIN / *INTEREST,ACTIVITIES,GAMES,HOBBIES,CLUB MEMBERSHIP ETC*

MINAT / HOBI (*INTERESTS/HOBBIES*)	AKTIVITI (*ACTIVITIES*)	KEAHLIAN KELAB (*CLUB MEMBERSHIP*)

KESEDIAAN / *AVAILABILITY*

GAJI DAN FAEDAH - FAEDAH LAIN YANG DIJANGKAKAN
SALARY AND OTHER BENEFITS EXPECTED

JIKA PERMOHONAN DITERIMA, NYATAKAN TARIKH ANDA BOLEH MULA BEKERJA?
IF APPLICATION IS ACCEPTED, WHEN CAN YOU START WORK?

NAMA JIKA BERLAKU KECEMASAN
EMERGENCY CONTACT PERSON
NAMA PENUH / *FULL NAME* HUBUNGAN / *RELATIONSHIP*

ALAMAT SURAT MENYURAT / *MAILING ADDRESS*
POSKOD / *POSTCODE* NEGERI / *STATE*
TEL (RUMAH / *HOUSE*) TB / *HP*

PENGAKUAN / *DECLARATION* SILA ☑ DI TEMPAT YANG BERKENAAN / *PLEASE* ☑ *WHERE IT IS APPICABLE*

1) ADAKAH ANDA MEMPUNYAI SEBARANG REKOD, JENAYAH / KES MAHKAMAH TERTANGGUH?
DO YOU HAVE ANY CRIMINAL RECORD / PENDING COURT CASE ?
JIKA YA, SILA JELASKAN (*IF YES, PLEASE EXPLAIN*) ☐ YA *YES* ☐ TIDAK *NO*

2) PERNAHKAN ANDA DISYTHARKAN BANKRAP DI MANA-MANA BANK / INSTITUSI KEWANGAN DI MANA-MANA NEGARA?
HAVE YOU BEEN DECLARED FOR BANKCRUPCY IN ANY BANK/FINANCIAL INSTITUTION IN ANY COUNTRY?
JIKA YA, SILA JELASKAN (*IF YES, PLEASE EXPLAIN*) ☐ YA *YES* ☐ TIDAK *NO*

3) PERNAHKAN ANDA DIBUANG / DIBERHENTIKAN KERJA SEBELUM INI?
HAVE YOU EVER BEEN DISMISSED / TERMINATED FROM EMPLOYMENT BEFORE?
JIKA YA, SILA JELASKAN (*IF YES, PLEASE EXPLAIN*) ☐ YA *YES* ☐ TIDAK *NO*

4) ADAKAH ANDA TERIKAT DENGAN MANA-MANA ORGANISASI?
ARE YOU BONDED BY ANY ORGANISATION?
JIKA YA, SILA JELASKAN (*IF YES, PLEASE EXPLAIN*) ☐ YA *YES* ☐ TIDAK *NO*

5 Part 3

5.1 1st File

This query acts as a " High-Value Procurement Cash Flow & Approval Tracker " for major capital expenditures.

It identifies the official name of the purchase order (to specify exactly which major asset or system is being acquired) and the subtotal amount from the corresponding invoice (to track the exactly financial liability before taxes and additional fees).

This specifically for 'APPROVED' purchase orders where the invoiced subtotal exceeds $900,000, effectively filtering for "Mega-Purchases" that require top-tier executive oversight and treasury planning.

```
SET LINESIZE 150
COLUMN purchase_order_name FORMAT A50
SET COLSEP ' | '
COLUMN SUBTOTAL_AMOUNT FORMAT 99,999,999

SELECT
    po.purchase_order_name,
    i.subtotal_amount
FROM
    purchase_order po
JOIN invoice i
    ON po.purchase_id = i.purchase_id
   AND po.purchase_reference_num = i.purchase_reference_num
WHERE po.order_approval_status = 'APPROVED'
  AND i.subtotal_amount > 900000
ORDER BY i.invoice_date;
```

Output:-

```
SQL> @C:\Users\User\OneDrive\Documents\Finalized\file1_cmd.sql

PURCHASE_ORDER_NAME                                | SUBTOTAL_AMOUNT
-------------------------------------------------- | ---------------
Radar Calibration System Order                     |       1,500,000
Precision Cutting Equipment Order                  |       1,200,000
Marine HVAC & Cooling System Order                 |       2,100,000
Marine Engine Diagnostic System Order              |         950,000
Navigation Testing System Order                    |       2,500,000
Propulsion Balancing System Order                  |       1,100,000

6 rows selected.
```

5.2 2nd File

Scenario :-

This query serves as a “Delivered Equipment Inventory & Cost Summary” for projects.

It identifies the equipment type (to categorize each major asset, such as Radar Calibration Systems or Population Balancing Systems), the total count of equipment (to quantify how many units of each type have been received), and the total cumulative cost (to calculate the exact financial investment in each equipment category).

This is specifically for purchase orders that have been ‘APPROVED’, deliveries that are already ‘Delivered’ (meaning the goods have physically arrived at the facility), and equipment that is in ‘Good’ condition (filtering out any damaged or defective items that would require returns or warranty claims). The results are grouped by equipment type.

```
SET LINESIZE 200
SET COLSEP ' | '
COLUMN equipment_type FORMAT A50
COLUMN total_equipment FORMAT 99,999,999
COLUMN total_cost FORMAT 99,999,999

SELECT
    e.equipment_type,
    COUNT(e.equipment_id) AS total_equipment,
    SUM(i.unit_cost) AS total_cost
FROM purchase_order po
JOIN delivery d
    ON po.purchase_id = d.purchase_id
   AND po.purchase_reference_num = d.purchase_reference_num
JOIN equipment e
    ON d.delivery_id = e.delivery_id
   AND d.tracking_number = e.tracking_number
JOIN inventory i
    ON e.inventory_item_id = i.inventory_item_id
   AND e.inventory_reference_number = i.inventory_reference_number
WHERE po.order_approval_status = 'APPROVED'
```

```
  AND d.delivery_status = 'Delivered'
  AND e.equipment_condition = 'Good'
GROUP BY e.equipment_type;
```

Output:-

```
SQL> @C:\Users\User\OneDrive\Documents\Finalized\file2_cmd.sql

EQUIPMENT_TYPE                                | TOTAL_EQUIPMENT |  TOTAL_COST
--------------------------------------------- | --------------- | -----------
Precision Cutting Equipment                   |               1 |      32,000
Radar Calibration System                      |               1 |     120,000
Automated Cable Installation System           |               1 |      54,000
Propulsion Balancing System                   |               1 |      77,000
Fuel System Diagnostic Equipment              |               1 |      45,000
```

5.3 3rd File

Scenario :-

The query serves as a comprehensive end-to-end operational deployment map, tracing the entire lifecycle of high-priority projects from the initial client request all the way down to the specific vessel and the employees assigned to it.

It identifies the client's ownership type and industry sector (to prioritize Government-Linked Companies and Government Agencies in the defence space), the service request title and requested service type (to understand the specific client need, such as submarine support or ship repairs), the contract title and contract scope (to define the exact legal and technical obligations agreed upon), the project name and project type (to categorize the initiative as Defence, Maintenance, or Upgrade), the vessel name and vessel category (to pinpoint exactly which naval asset from Submarine Support Vessels to Frigates is being mobilized), and finally the employee name alongside their job position (to verify which human resources are physically tied to that specific vessel).

This is specifically for 'Active' clients with service requests flagged as 'High' priority, contracts that were signed after 15th May 2026, and projects currently 'In Progress'. The results are ordered by project start date (ascending).

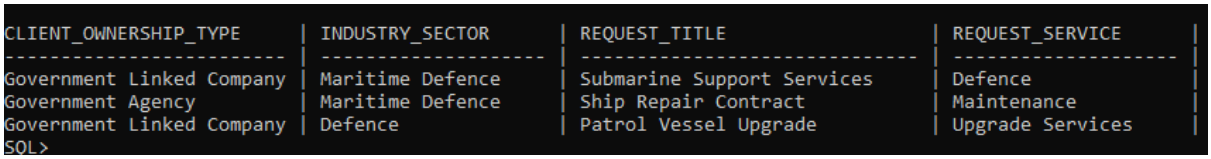
```
SET LINESIZE 500
SET WRAP OFF
SET TRIMOUT ON
SET TRIMSPOOL ON
SET COLSEP ' | '
SET PAGESIZE 100
SET FEEDBACK OFF

COLUMN CLIENT_OWNERSHIP_TYPE     FORMAT A25
COLUMN INDUSTRY_SECTOR           FORMAT A20
COLUMN REQUEST_TITLE             FORMAT A30
COLUMN REQUEST_SERVICE           FORMAT A20
COLUMN CONTRACT_TITLE            FORMAT A35
COLUMN CONTRACT_SCOPE            FORMAT A45
COLUMN PROJECT_NAME              FORMAT A30
COLUMN PROJECT_TYPE              FORMAT A15
```

```
COLUMN VESSEL_NAME                FORMAT A20
COLUMN VESSEL_CATEGORY            FORMAT A25
COLUMN EMPLOYEE_NAME              FORMAT A15
COLUMN EMPLOYEE_JOB_POSITION      FORMAT A25
SELECT
    c.client_ownership_type,
    c.industry_sector,
    sr.request_title,
    sr.request_service,
    ct.contract_title,
    ct.contract_scope,
    p.project_name,
    p.project_type,
    v.vessel_name,
    v.vessel_category,
    e.employee_name,
    e.employee_job_position
FROM client c
JOIN service_request sr
    ON c.client_id = sr.client_id
   AND c.client_reference_num = sr.client_reference_num
JOIN contract ct
    ON sr.request_id = ct.request_id
   AND sr.request_code = ct.request_code
JOIN project p
    ON ct.contract_id = p.contract_id
   AND ct.agreement_reference_number = p.agreement_reference_number
JOIN project_vessel pv
    ON p.project_id = pv.project_id
   AND p.project_reference_num = pv.project_reference_num
JOIN vessel v
    ON pv.vessel_id = v.vessel_id
```

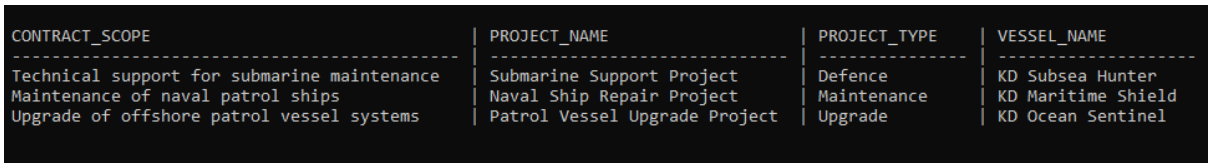
```
   AND pv.vessel_registration_number = v.vessel_registration_number
JOIN vessel_employee ve
    ON v.vessel_id = ve.vessel_id
   AND v.vessel_registration_number = ve.vessel_registration_number
JOIN employee e
    ON ve.employee_id = e.employee_id
   AND ve.employee_nric = e.employee_nric
WHERE c.client_status = 'Active'
  AND sr.request_priority_level = 'High'
  AND ct.contract_sign_date > DATE '2026-05-15'
  AND p.project_status = 'In Progress'
ORDER BY p.project_start_date;
```

Output :-

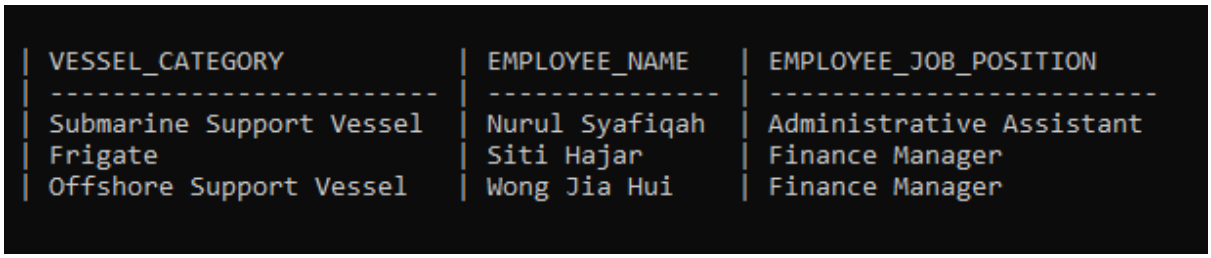


CLIENT_OWNERSHIP_TYPE	INDUSTRY_SECTOR	REQUEST_TITLE	REQUEST_SERVICE
Government Linked Company	Maritime Defence	Submarine Support Services	Defence
Government Agency	Maritime Defence	Ship Repair Contract	Maintenance
Government Linked Company	Defence	Patrol Vessel Upgrade	Upgrade Services

SQL>



CONTRACT_SCOPE	PROJECT_NAME	PROJECT_TYPE	VESSEL_NAME
Technical support for submarine maintenance	Submarine Support Project	Defence	KD Subsea Hunter
Maintenance of naval patrol ships	Naval Ship Repair Project	Maintenance	KD Maritime Shield
Upgrade of offshore patrol vessel systems	Patrol Vessel Upgrade Project	Upgrade	KD Ocean Sentinel



VESSEL_CATEGORY	EMPLOYEE_NAME	EMPLOYEE_JOB_POSITION
Submarine Support Vessel	Nurul Syafiqah	Administrative Assistant
Frigate	Siti Hajar	Finance Manager
Offshore Support Vessel	Wong Jia Hui	Finance Manager

5.4 4th File

Scenario :-

This query is to provide a high-level oversight summary of senior staff deployment and their attendance integrity for new, high-stakes engineering projects.

It identifies the ownership type of the client (to prioritize strategic partners), the specific project name and associated work order title (to define the exact scope of hazardous operations), the name of the senior employee assigned, their department's emergency contact number (for immediate escalation), a distinct summary of their attendance statuses (to flag any deviations from perfect attendance), their gross salary (to justify the cost of senior manpower), and the hazard category of the materials being handled (to assess risk levels).

This is specifically for 'Active' clients whose new projects started on or after 10th May 2026, involving only 'Active' senior employees earning a gross salary of RM5,000 or more. The query targets attendance records logged within the same project timeframe (since 10th May 2026). The results are displayed in descending order of the client's ownership type (alphabetically).

```
SET LINESIZE 250
SET FEEDBACK OFF
SET COLSEP ' | '
SET TRIMOUT ON
SET TRIMSPOOL ON
SET WRAP ON

COLUMN CLIENT_OWNERSHIP_TYPE FORMAT A25
COLUMN PROJECT_NAME FORMAT A35
COLUMN WORK_ORDER_TITLE FORMAT A37
COLUMN EMPLOYEE_NAME FORMAT A15
COLUMN DEPARTMENT_CONTACT_NUMBER FORMAT A26
COLUMN ATTENDANCE_STATUS FORMAT A20
COLUMN GROSS_SALARY FORMAT 99,999
COLUMN MATERIAL_HAZARD_CATEGORY FORMAT A25
SELECT DISTINCT
    c.client_ownership_type,
```

```
    p.project_name,
    wo.work_order_title,
    e.employee_name,
    d.DEPARTMENT_CONTACT_NUMBER ,
    a.ATTENDANCE_STATUS   ,
    ps.gross_salary,
    m.MATERIAL_HAZARD_CATEGORY
FROM client c
JOIN project p
    ON c.client_id = p.client_id
    AND c.client_reference_num = p.client_reference_num
JOIN work_order wo
    ON p.project_id = wo.project_id
    AND p.project_reference_num = wo.project_reference_num
JOIN employee_project ep
    ON p.project_id = ep.project_id
    AND p.project_reference_num = ep.project_reference_num
JOIN employee e
    ON ep.employee_id = e.employee_id
    AND ep.employee_nric = e.employee_nric
JOIN department d
    ON e.department_id = d.department_id
    AND e.department_code = d.department_code
JOIN attendance a
    ON e.employee_id = a.employee_id
    AND e.employee_nric = a.employee_nric
JOIN payslip ps
    ON e.employee_id = ps.employee_id
    AND e.employee_nric = ps.employee_nric
JOIN material_work_order mwo
    ON wo.work_order_id = mwo.work_order_id
    AND wo.work_order_num = mwo.work_order_num
```

```
JOIN material m
    ON mwo.material_item_id = m.material_item_id
    AND mwo.material_code = m.material_code
WHERE c.client_status = 'Active'
  AND ps.gross_salary >= 5000
  AND e.employment_status = 'Active'
  AND p.project_start_date >= DATE '2026-05-10'
  AND a.ATTENDANCE_DATE  >= DATE '2026-05-10'
ORDER BY c.CLIENT_OWNERSHIP_TYPE  DESC;
```

Output :-

```
SQL> @C:\Users\User\OneDrive\Documents\Finalized\file4_cmd.sql

CLIENT_OWNERSHIP_TYPE     | PROJECT_NAME                      | WORK_ORDER_TITLE                      | EMPLOYEE_NAME   |
------------------------- | --------------------------------- | ------------------------------------- | --------------- |
Private Limited           | Radar Integration Project         | Radar Integration Work Order          | Lim Wei Jian    |
Government Linked Company | OPV Construction Project          | Vessel Construction Work Order        | Ahmad Razak     |
Government Agency         | Fleet Maintenance Planning Project | Fleet Maintenance Planning Work Order | Nur Aisyah      |
Consortium                | Engine Overhaul Project           | Engine Overhaul Work Order            | Mohd Faiz       |
```

```
DEPARTMENT_CONTACT_NUMBER | ATTENDANCE_STATUS    | GROSS_SALARY | MATERIAL_HAZARD_CATEGORY
------------------------- | -------------------- | ------------ | ------------------------
03-7788 1003              | Present              |        6,500 | High
03-7788 1001              | Present              |        8,500 | Low
03-7788 1004              | Present              |        5,500 | Low
03-7788 1005              | Present              |        5,000 | Low
```

5.5 5th File

Scenario :-

This view serves as a "Recent Contract Activity Summary Dashboard" providing a high-level, readable snapshot of all newly signed contractual agreements with active clients in the defence and maritime sector.

It identifies the client ID, request ID, and contract ID (to maintain precise referential integrity across the procurement lifecycle), a concatenated client information field that combines the `client_ownership_type` (such as Government Linked Company, Government Agency, Consortium, or Private Limited) with their `client_loyalty` tier (Platinum, Gold, or Bronze) to provide immediate strategic context about the client's value and relationship status, and a concatenated request information field that merges the `request_title` with the `request_service` type (such as Shipbuilding, Defence, Maintenance, Upgrade Services, or Technology) to quickly communicate exactly what service is being contracted.

This is specifically for contracts with a contract sign date after 10th May 2026, effectively filtering for freshly inked deals that have entered the execution pipeline within the last two months. The results are ordered chronologically by contract sign date, giving management a clear timeline of when these new commitments came on board (ascending).

```
SET LINESIZE 200
SET COLSEP ' | '
COLUMN CLIENT_ID        FORMAT A15
COLUMN REQUEST_ID       FORMAT A15
COLUMN CONTRACT_ID      FORMAT A15
COLUMN CLIENT_INFO      FORMAT A40
COLUMN REQUEST_INFO     FORMAT A50

CREATE OR REPLACE VIEW recent_contracts_view AS
SELECT
    c.client_id,
    sr.request_id,
    ct.contract_id,
    c.client_ownership_type || ' - ' || c.client_loyalty AS
client_info,
```

```
    sr.request_title || ' - ' || sr.request_service AS request_info
FROM client c
JOIN service_request sr
    ON c.client_id = sr.client_id
   AND c.client_reference_num = sr.client_reference_num
JOIN contract ct
    ON sr.request_id = ct.request_id
   AND sr.request_code = ct.request_code
WHERE ct.contract_sign_date > DATE '2026-05-10'
ORDER BY ct.contract_sign_date;
SELECT * FROM recent_contracts_view;
```

Output :-

```
SQL> @C:\Users\User\OneDrive\Documents\Finalized\file5_cmd.sql

CLIENT_ID       | REQUEST_ID      | CONTRACT_ID     | CLIENT_INFO                                | REQUEST_INFO
--------------- | --------------- | --------------- | ------------------------------------------ | ----------------------------------------------------
C001            | R001            | CT001           | Government Linked Company - Platinum       | Naval Vessel Construction Request - Shipbuilding
C008            | R008            | CT008           | Government Linked Company - Platinum       | Submarine Support Services - Defence
C005            | R005            | CT005           | Consortium - Gold                          | Engine Overhaul Project - Engine Services
C002            | R002            | CT002           | Government Agency - Gold                   | Ship Repair Contract - Maintenance
C004            | R004            | CT004           | Government Linked Company - Platinum       | Patrol Vessel Upgrade - Upgrade Services
C007            | R007            | CT007           | Private Limited - Bronze                   | Radar System Integration - Technology
C010            | R010            | CT010           | Government Agency - Platinum               | Fleet Maintenance Planning - Maritime Security
```

--- The End of Report ---